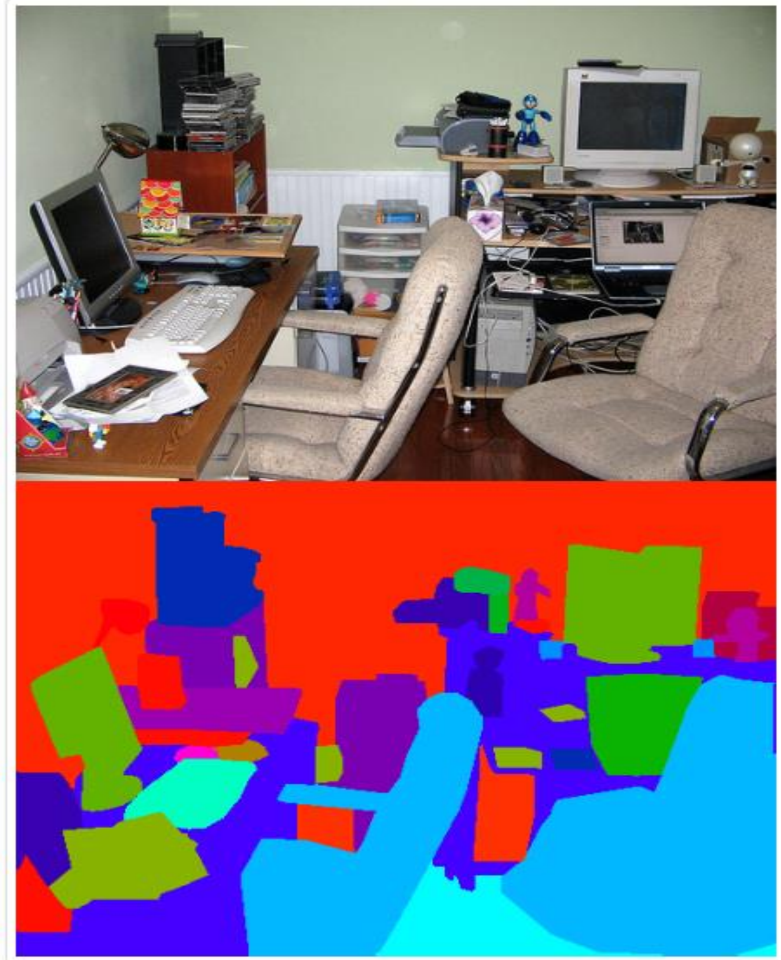
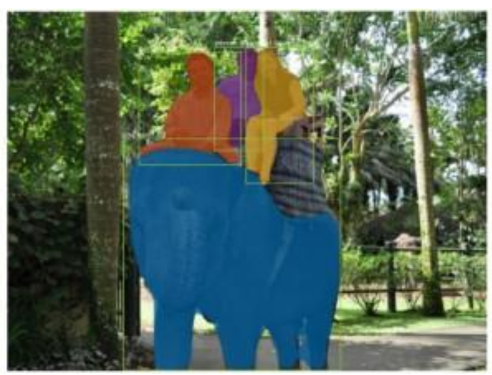
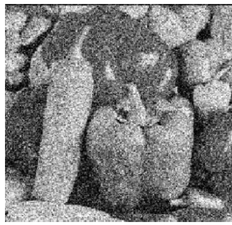
Les 7: Semantic Image segmentation





Computer vision applications

Image Enhancement



image

Classification

“dog”

label

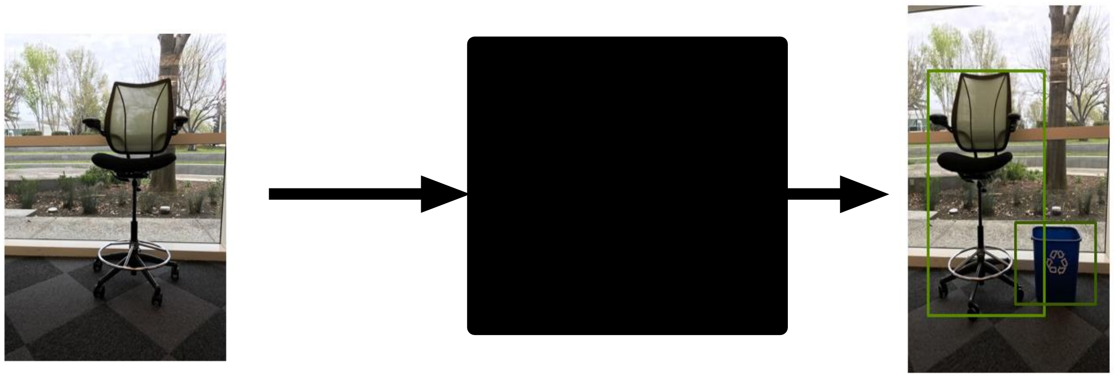
Embedding / Metric Learning

“45A4CE12006”

description vector

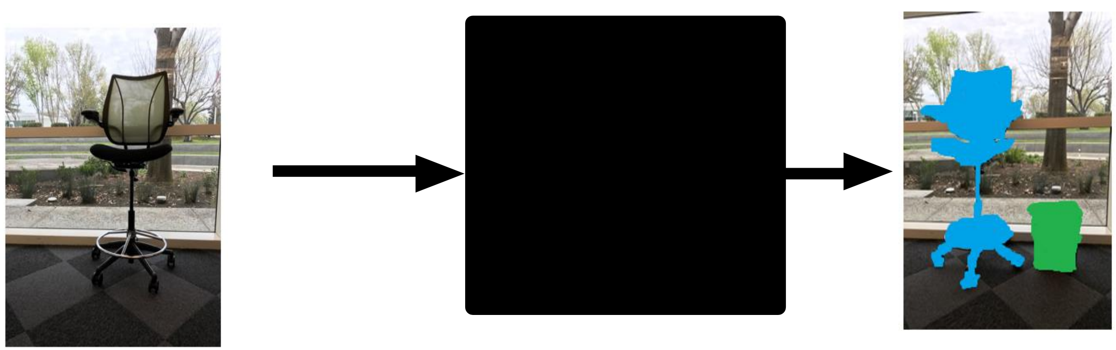
Computer vision applications

Object detection



labeled bounding boxes

Semantic Segmentation

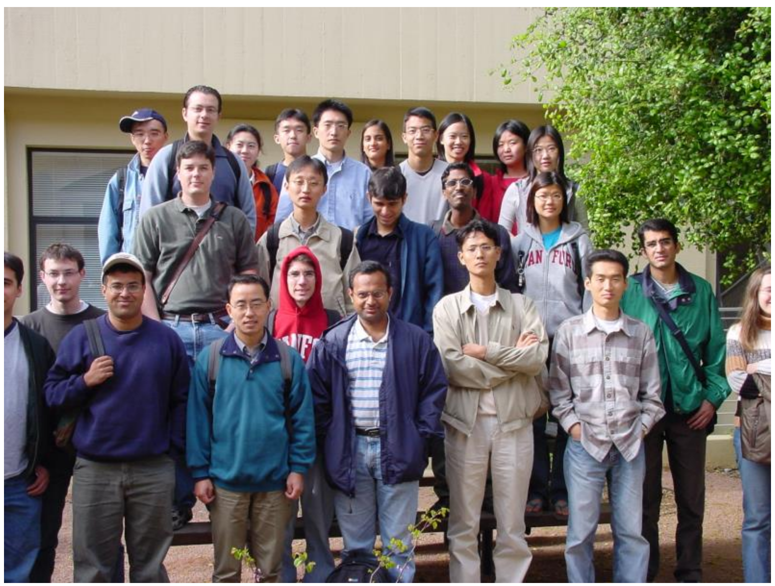


labeled pixels

Semantic Image Segmentation

- Semantic segmentation:
  - Classic techniques using colour
  - Fully Convolutional Network
  - Deconvolution
  - Skip connections: U-Net
- Instance Segmentation
  - Two-stage: Mask-RCNN
  - One-stage: YOLACT

SKIN COLOR SEGMENTATION



INPUT IMAGE

BINARY MASK

SKIN COLOR SEGMENTATION

BINARY MASK

BINARY MASK AFTER PROCESSING

Semantic Image Segmentation

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Semantic and instance segmentation

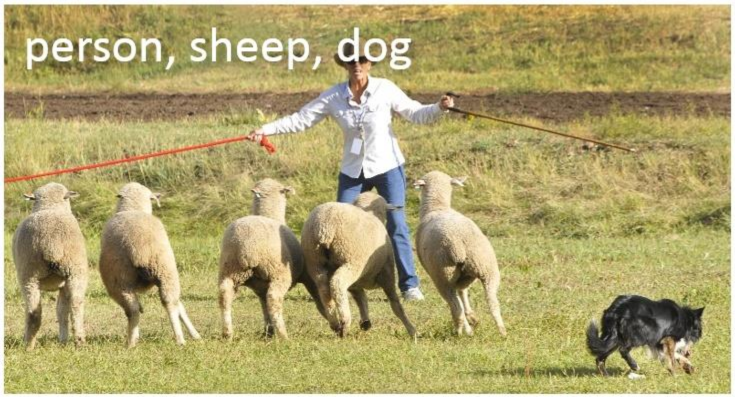
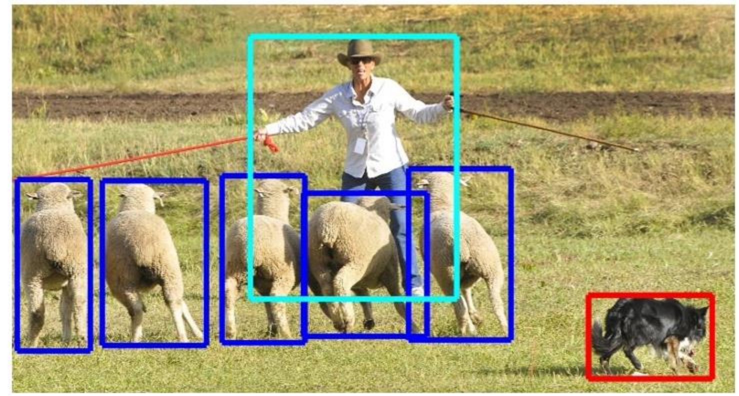
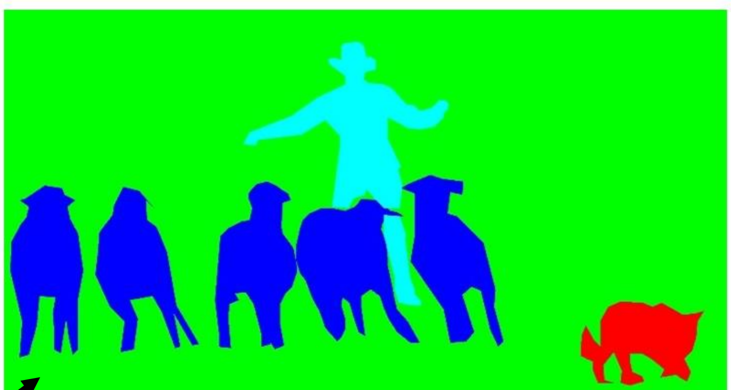


Image classification

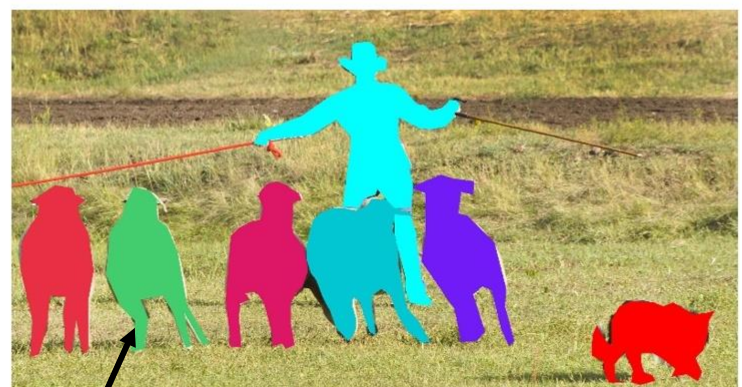


Object detection



Semantic segmentation

"stuff", each **pixel** is classified

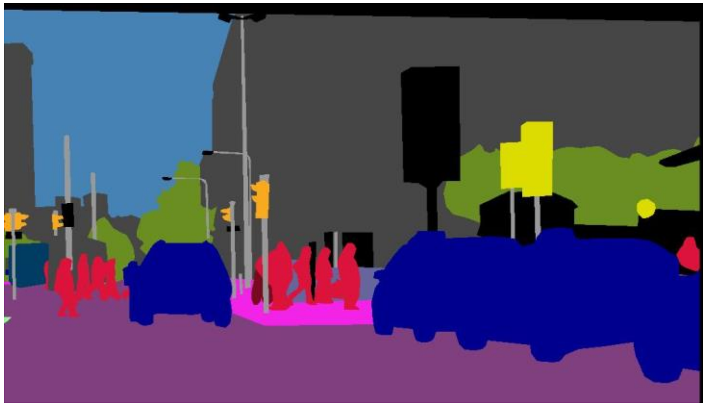


Instance segmentation

"things", each **instance** is classified

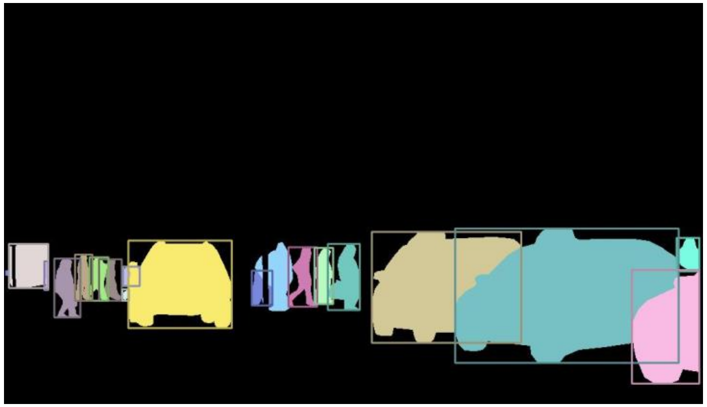
Semantic segmentation

- Each **pixel** is classified
- Output of network is tensor of shape **L x W x H**
  - With L the number of classes, and (W, H) the size of the input image
- E.g. U-Net
  - Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *ArXiv:1505.04597 [Cs]*. http://arxiv.org/abs/1505.04597



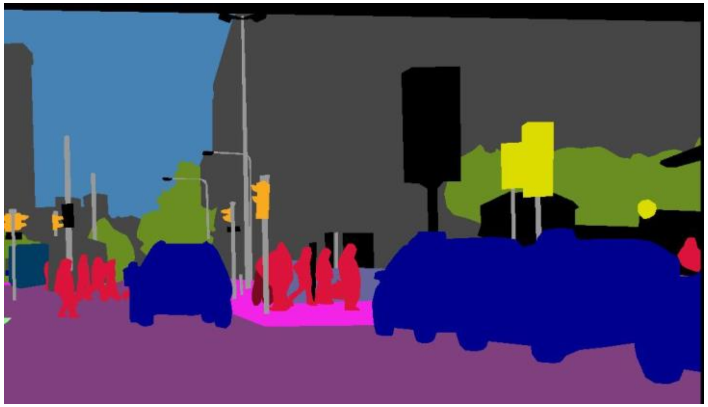
Instance segmentation

- Each **instance** is segmented and classified
  - Like object detection, but with segmentation masks instead of bounding boxes
- Output of the network is **a list of masks** with localization and classification predictions
- E.g. Mask R-CNN
  - He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask R-CNN. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2961–2969. https://doi.org/10.1109/TPAMI.2018.2844175

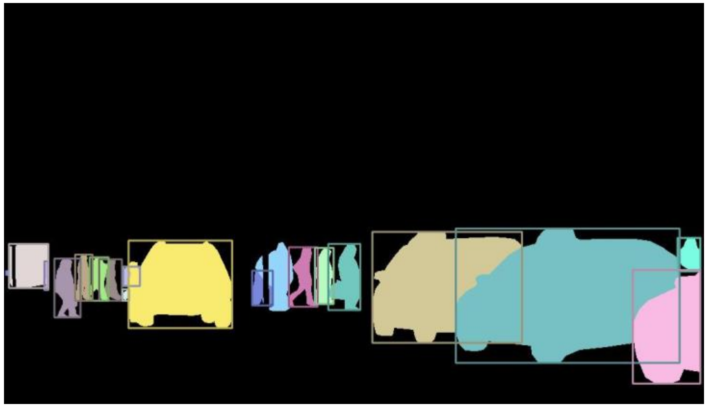


Extra: panoptic segmentation

(won't go into detail)



Semantic segmentation



Instance segmentation

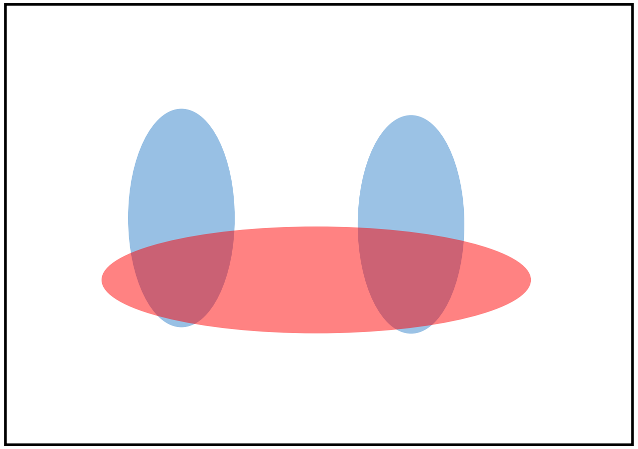
Panoptic segmentation

"stuff" *and* "things": **each pixel and each instance** is classified

Kirillov, A., He, K., Girshick, R., Rother, C., & Dollár, P. (2019). Panoptic Segmentation. *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 9404–9413.

Evaluation metric

- Pixel classification!
- Accuracy?
  - Heavily unbalanced
  - Common classes are over-emphasized
- *Intersection over Union*
  - Average across classes and images
- Per-class accuracy
  - Compute accuracy for every class and then average



Things vs Stuff

THINGS

- Person, cat, horse, etc
- Constrained shape
- Individual instances with separate identity
- May need to look at objects

STUFF

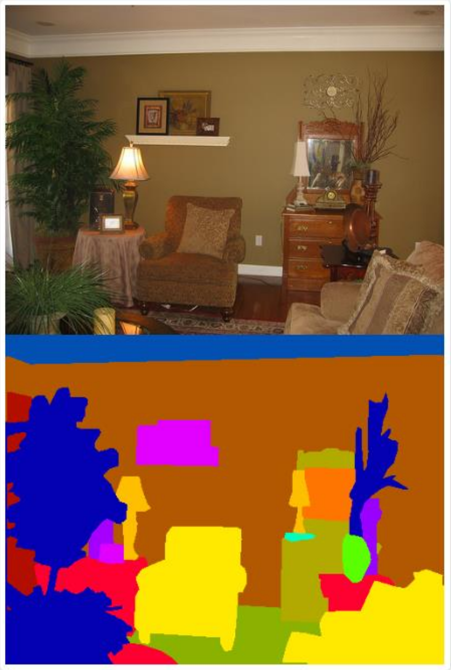
- Road, grass, sky etc
- Amorphous, no shape
- No notion of instances
- Can be done at pixel level
- “texture”

Challenges in data collection

- Precise localization is hard to annotate
- Annotating every pixel leads to heavy tails
- Common solution: annotate few classes (often things), mark rest as "Other"
- Common datasets: PASCAL VOC 2012 (~1500 images, 20 categories), COCO (~100k images, 20 categories)

Segmentation datasets

- Pascal Context
  - 10k+ training images with 400+ labels
  - semantic segmentation dataset


Segmentation datasets

- Microsoft Common Objects in COntext (COCO)
  - Originally: instance segmentation, but nowadays also contains annotations for stuff segmentation
  - 328k images with 91 labels

Semantic Image Segmentation

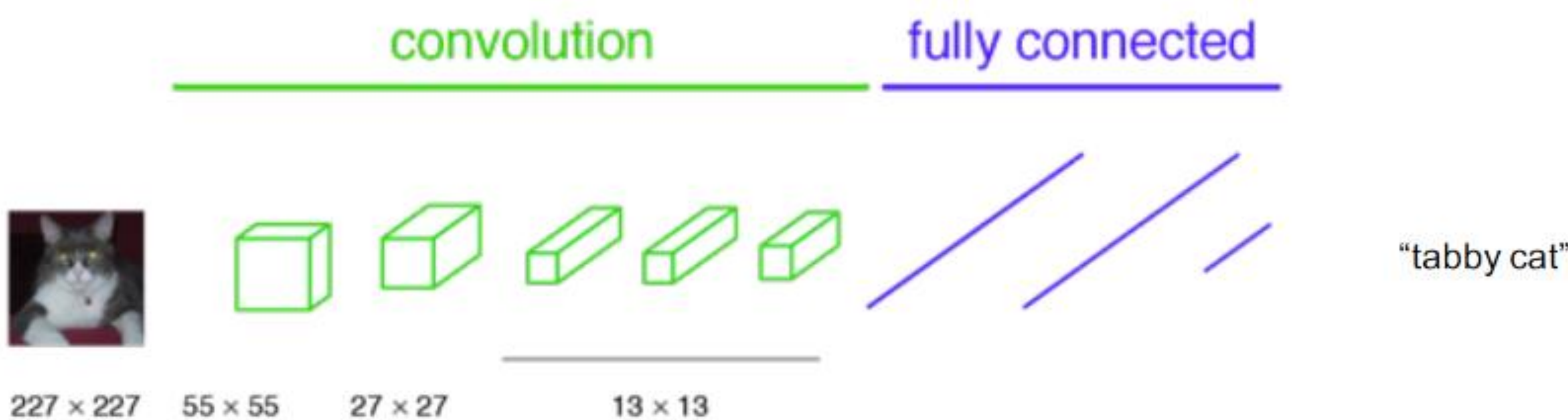
- Semantic segmentation:
  - Classic techniques using colour
  - Fully Convolutional Network
  - Deconvolution
  - Skip connections: U-Net
- Instance Segmentation
  - Two-stage: Mask-RCNN
  - One-stage: YOLACT

Fully Convolutional Network (FCN)

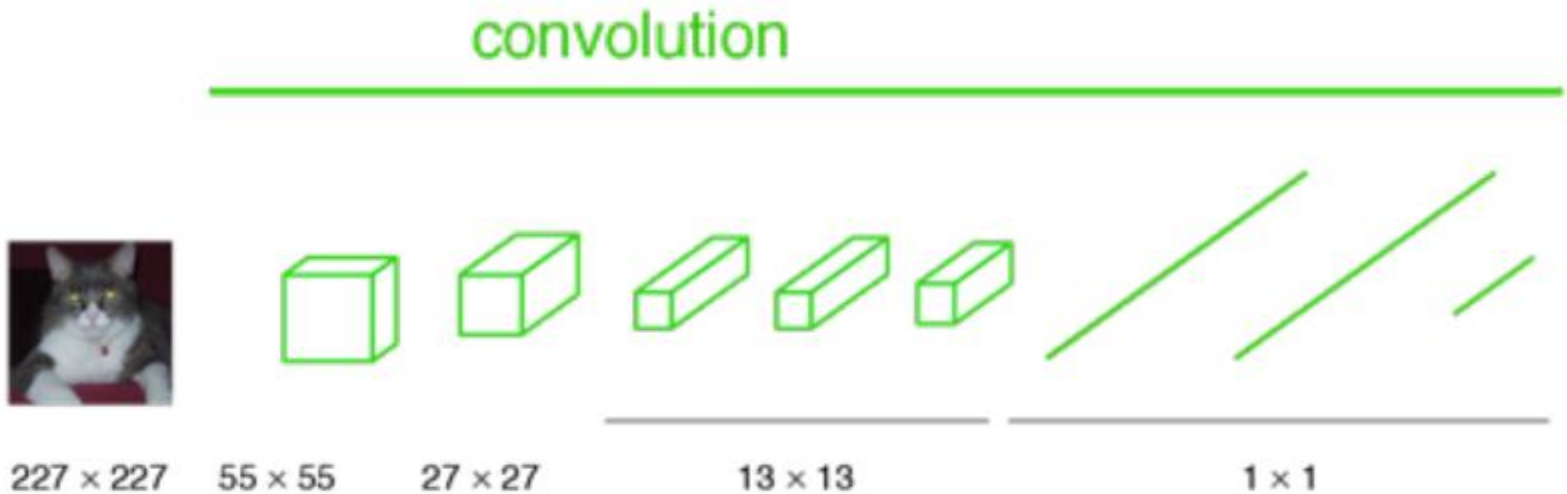
Long, J., Shelhamer, E., & Darrell, T. (2014). *Fully Convolutional Networks for Semantic Segmentation*. https://arxiv.org/abs/1411.4038v2

- Network for **semantic segmentation**
- Fully-connected layers in typical CNN classifiers are **replaced by conv layers**
  - Kernel size = size of fully-connected layer input
- Hence, network can take any input size
- Output feature maps are up-sampled to original input image size with **transposed convolutions**

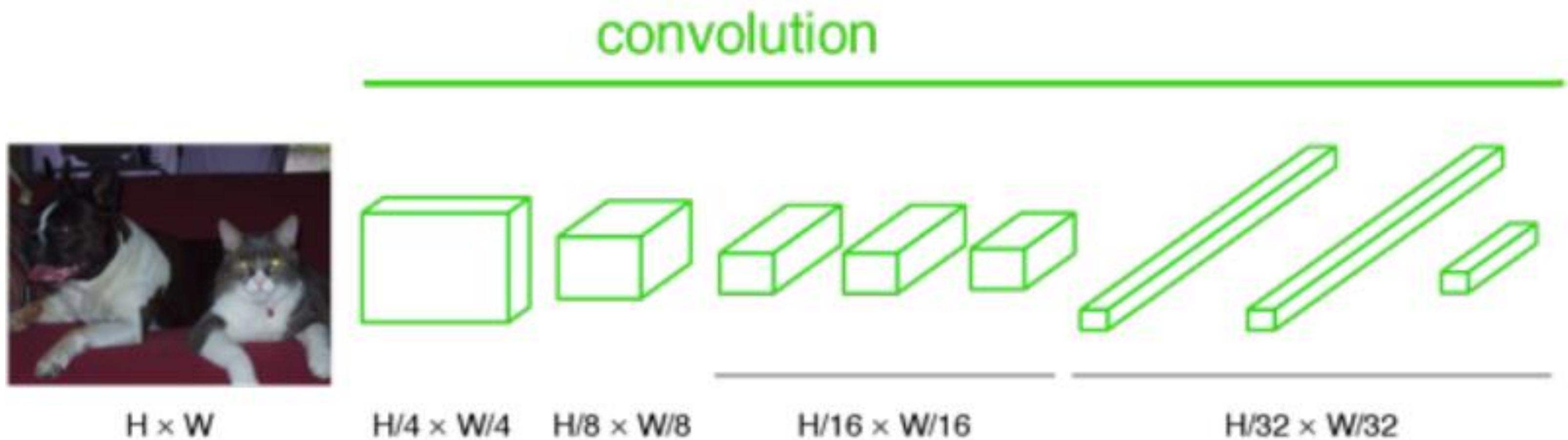
A typical CNN classifier



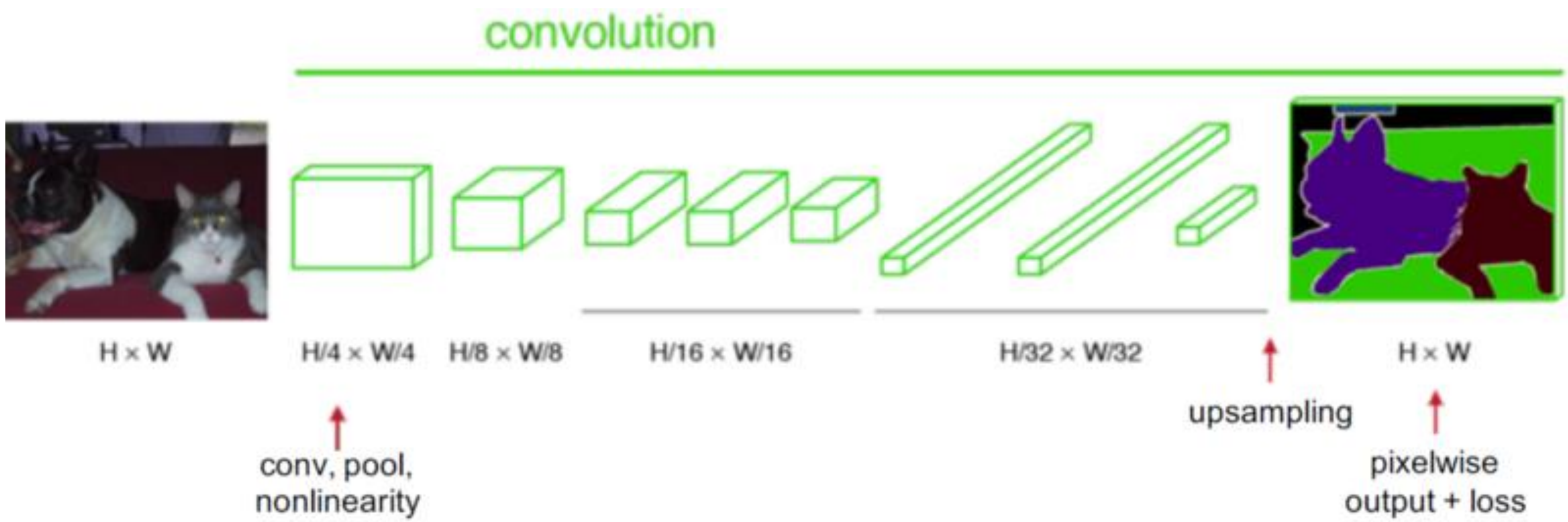
Replace fully-connected layers with convolutions



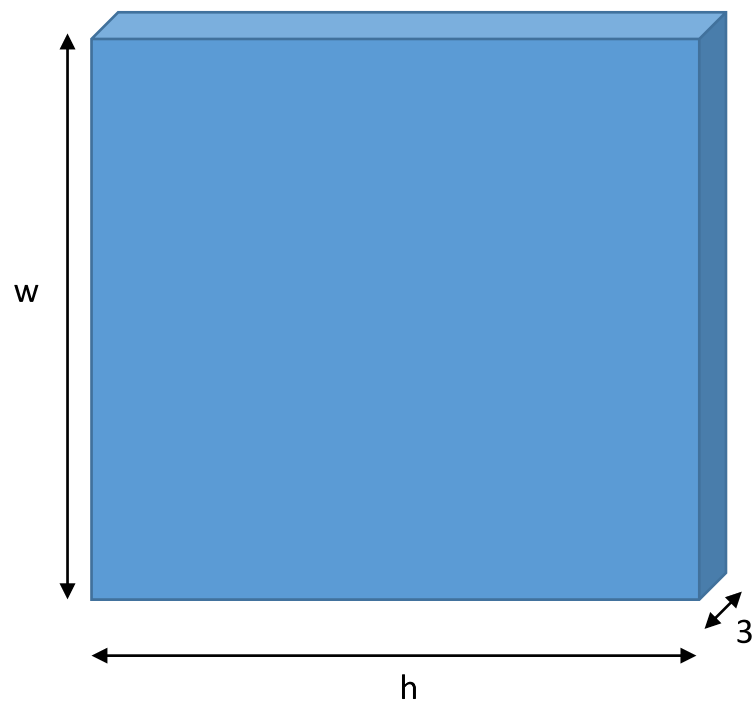
Network can take any input size!



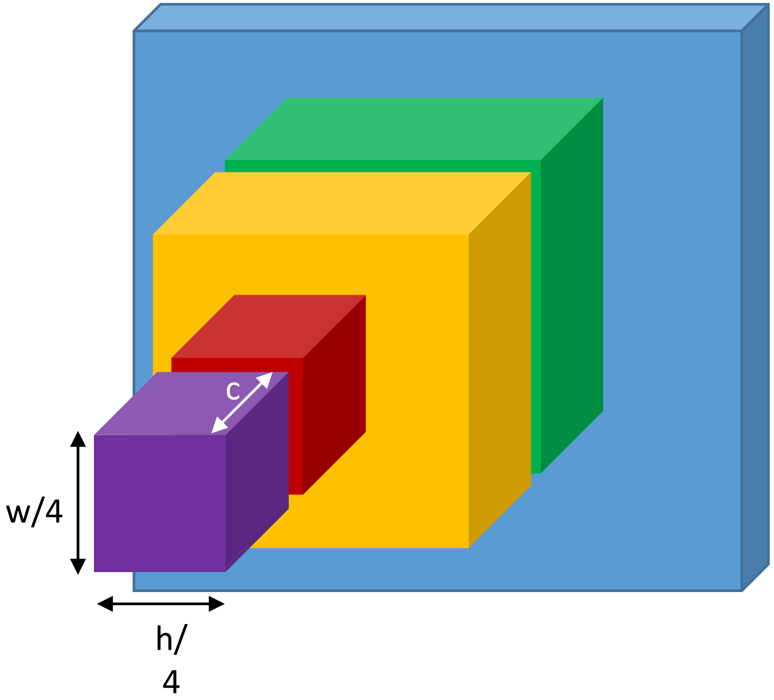
Upsample output feature maps



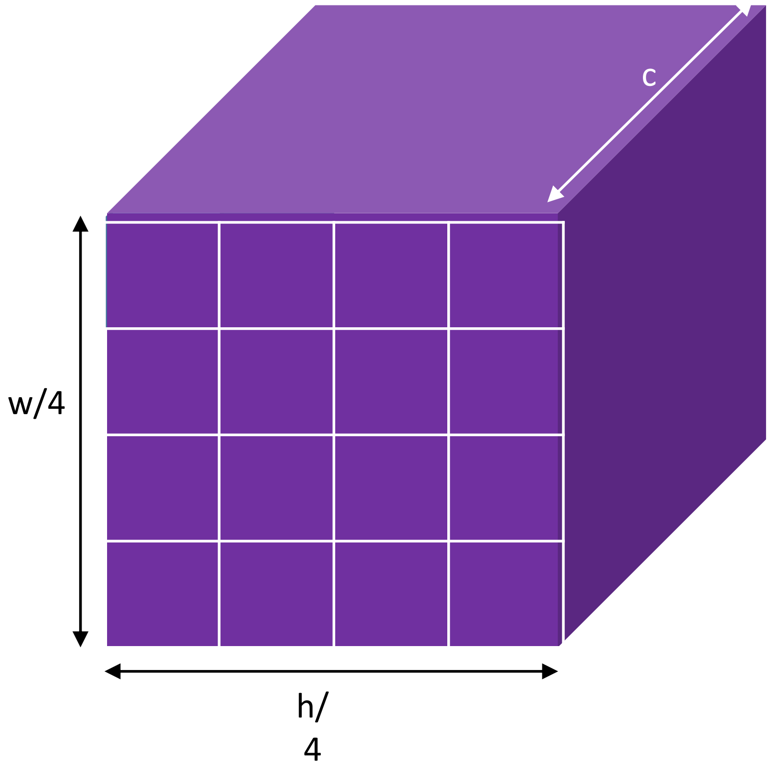
Semantic segmentation using convolutional networks



Semantic segmentation using convolutional networks

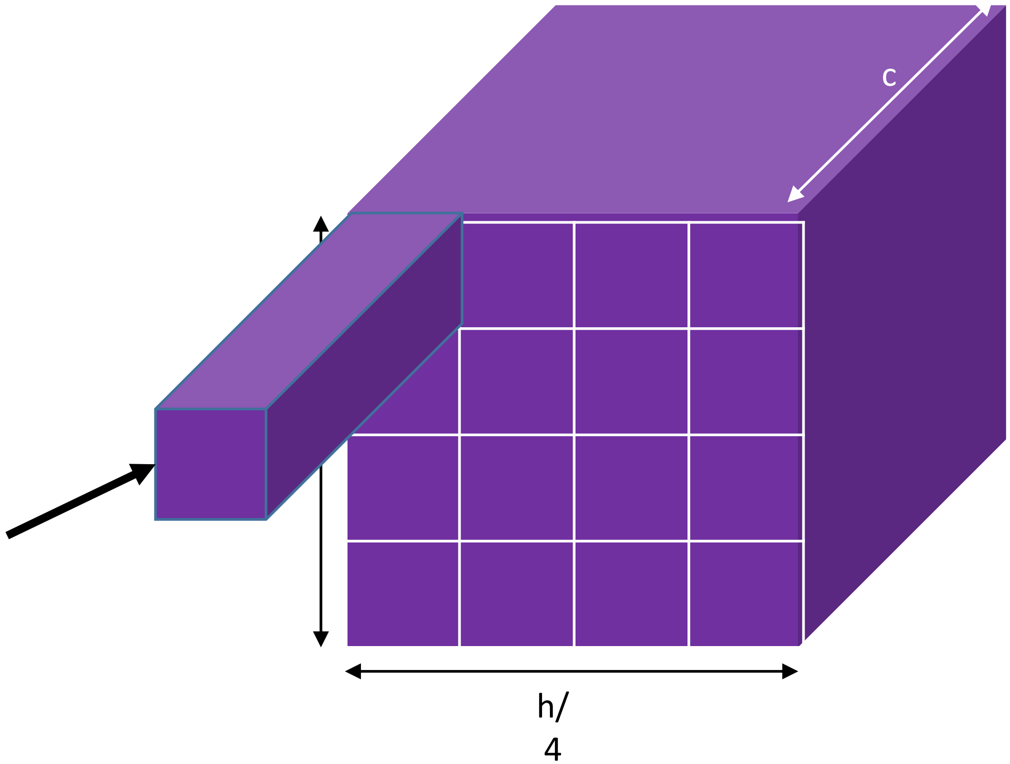


Semantic segmentation using convolutional networks

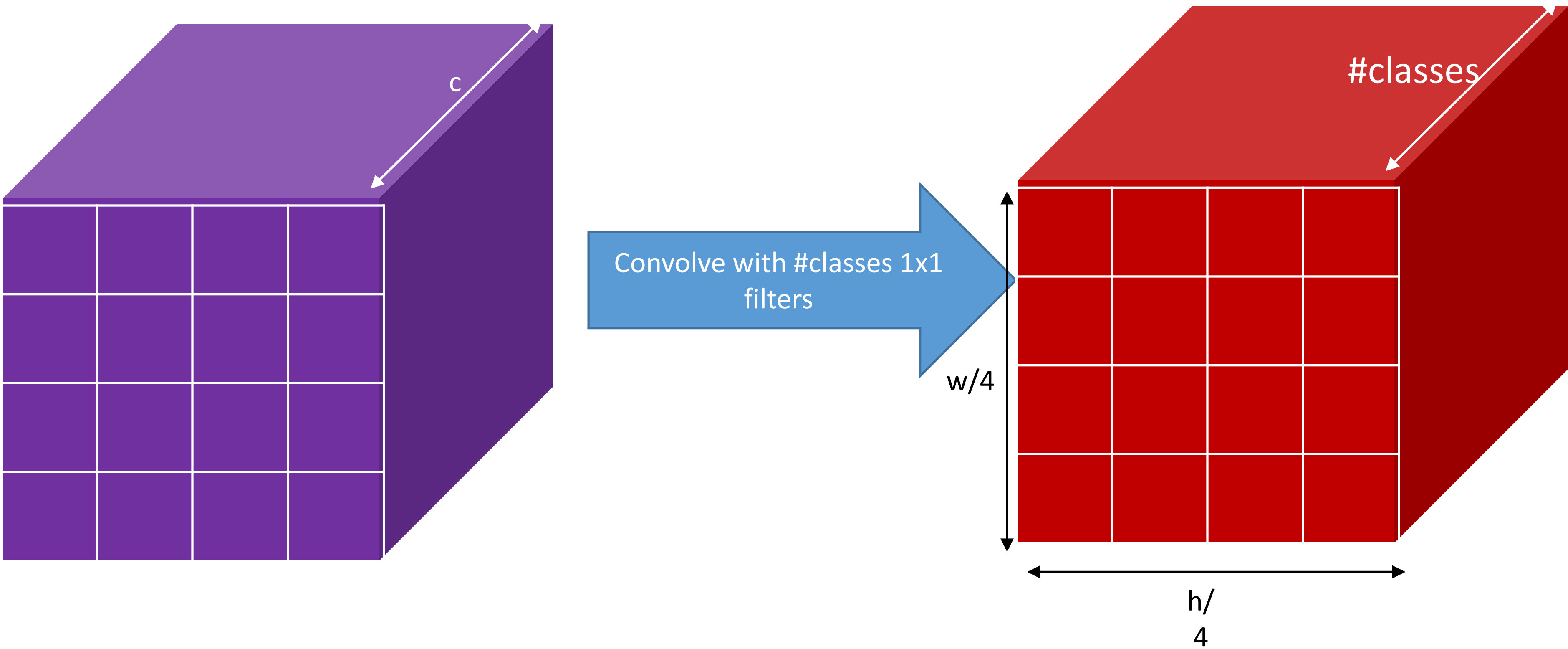


Semantic segmentation using convolutional networks

Can be considered as a feature vector for a pixel



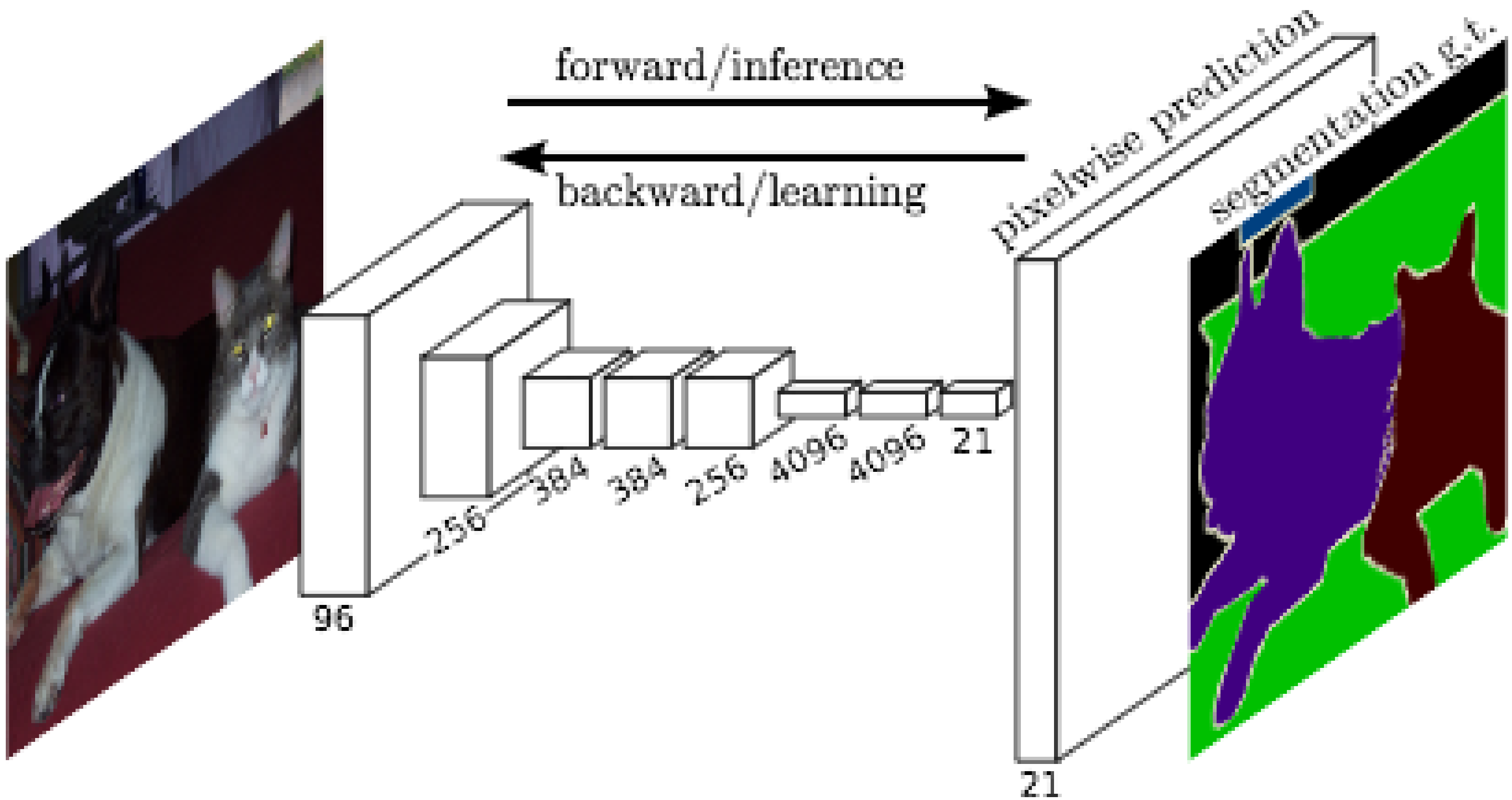
Semantic segmentation using convolutional networks



Semantic segmentation using convolutional networks

- Pass image through convolution and subsampling layers
- Final convolution with #classes outputs
- Get scores for *subsampled* image
- Upsample back to original size

Fully Convolutional Network (FCN)



Fully Convolutional Network

Semantic Image Segmentation

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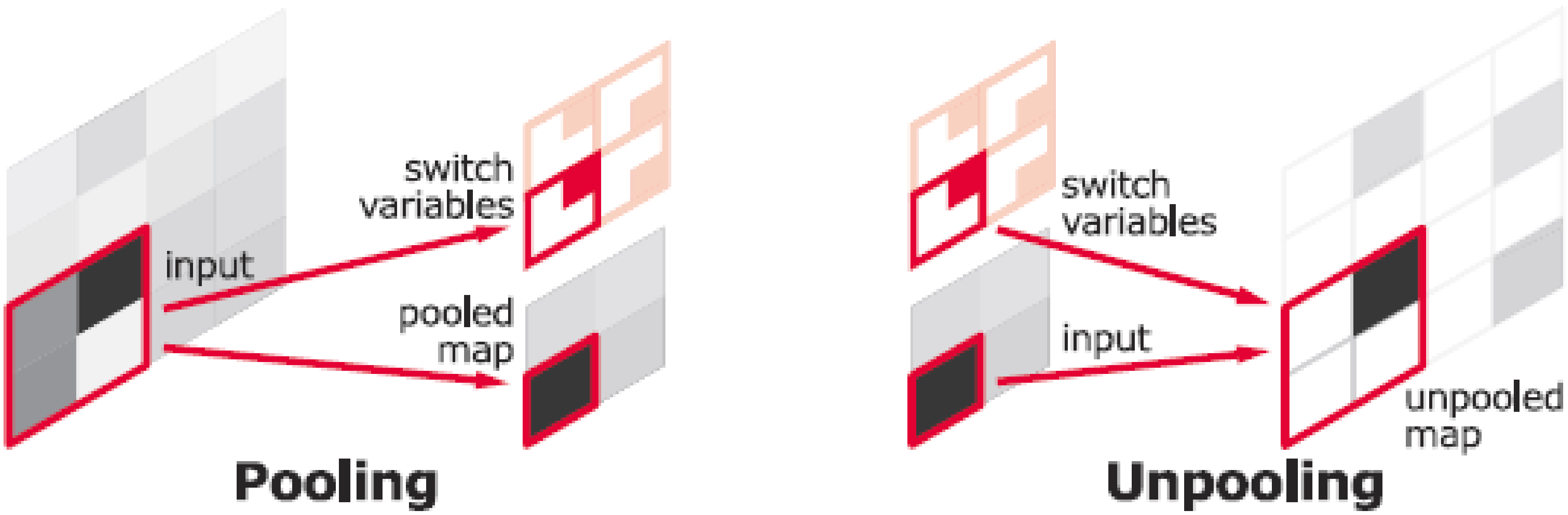
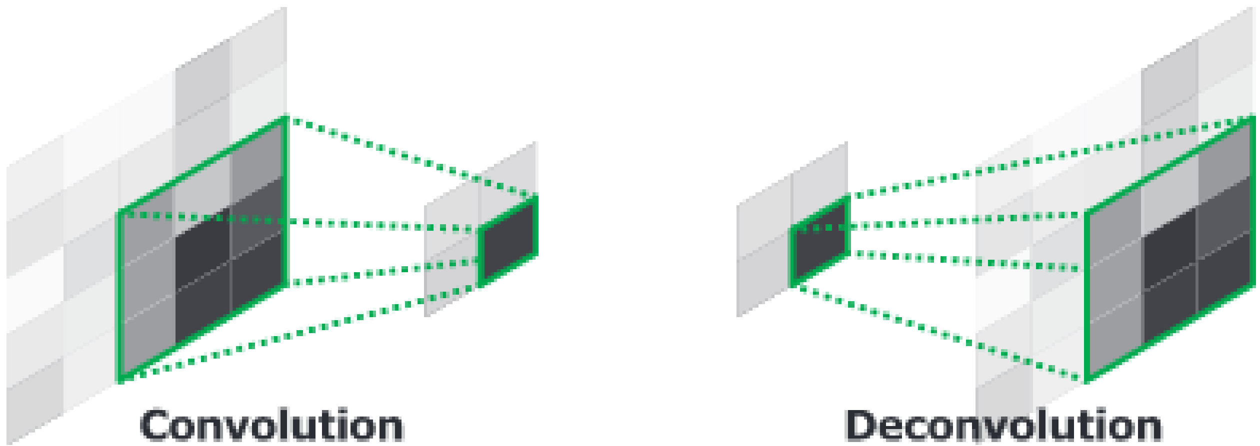
The resolution issue

- Problem: Need fine details!
- Shallower network / earlier layers?
  - Deeper networks work better: more abstract concepts
  - Shallower network => Not very semantic!
- Remove subsampling?
  - Subsampling allows later layers to capture larger and larger patterns
  - Without subsampling => Looks at only a small window!
- After subsampling: reverse operation = upsampling needed!

Reversing layers: Unpooling



Reversing layers: Deconvolution



Deconvolution = transposed convolution (=up-convolution)

- https://naokishibuya.medium.com/up-sampling-with-transposed-convolution-9ae4f2df52d0

Transposed convolutions

- Aka up-convolution, fractionally-strided convolution or deconvolution
- Implemented by swapping forward and backward of regular convolution
- Transposed conv with stride *s*, kernel size *k* and padding *p* will produce output size of

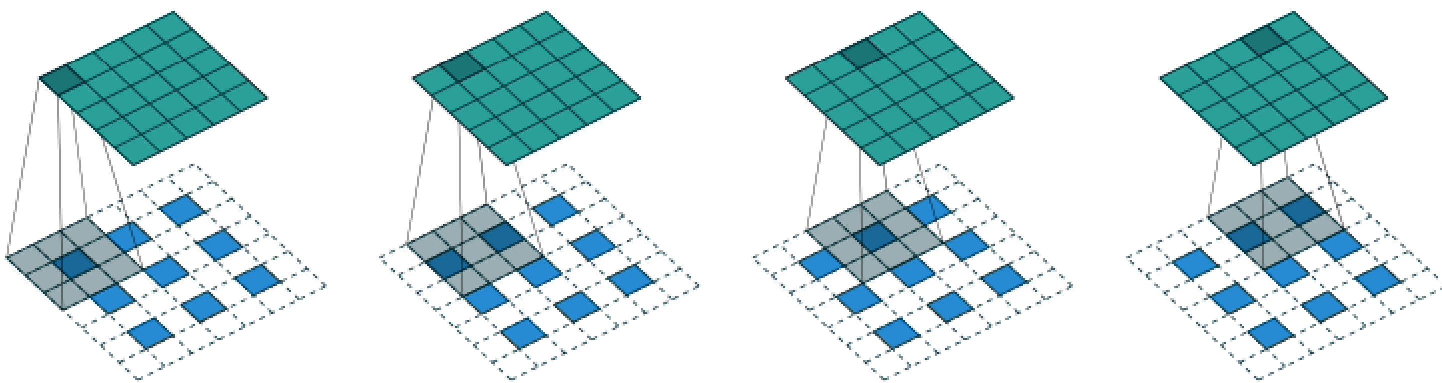
$$o' = s(i' - 1) + k - 2p$$

Dumoulin, V., & Visin, F. (2018). A guide to convolution arithmetic for deep learning. *ArXiv:1603.07285 [Cs, Stat]*. http://arxiv.org/abs/1603.07285

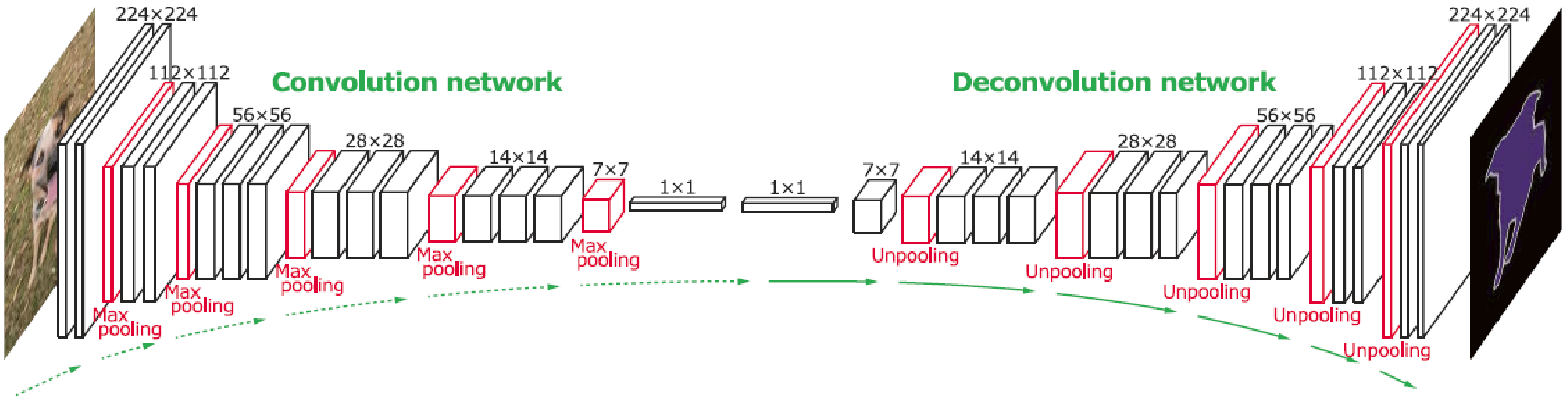
https://naokishibuya.medium.com/up-sampling-with-transposed-convolution-9ae4f2df52d0

Transposed convolutions

- E.g., *i' = 3, k = 3, s = 1, p = 1*



DeconvNet



224×224
112×112
Convolution network
56×56
28×28
14×14
7×7
1×1
1×1
7×7
14×14
Deconvolution network
28×28
56×56
112×112
224×224
Max pooling
Max pooling
Max pooling
Max pooling
Max pooling
Unpooling
Unpooling
Unpooling
Unpooling
Unpooling

Semantic Image Segmentation

- Semantic segmentation:
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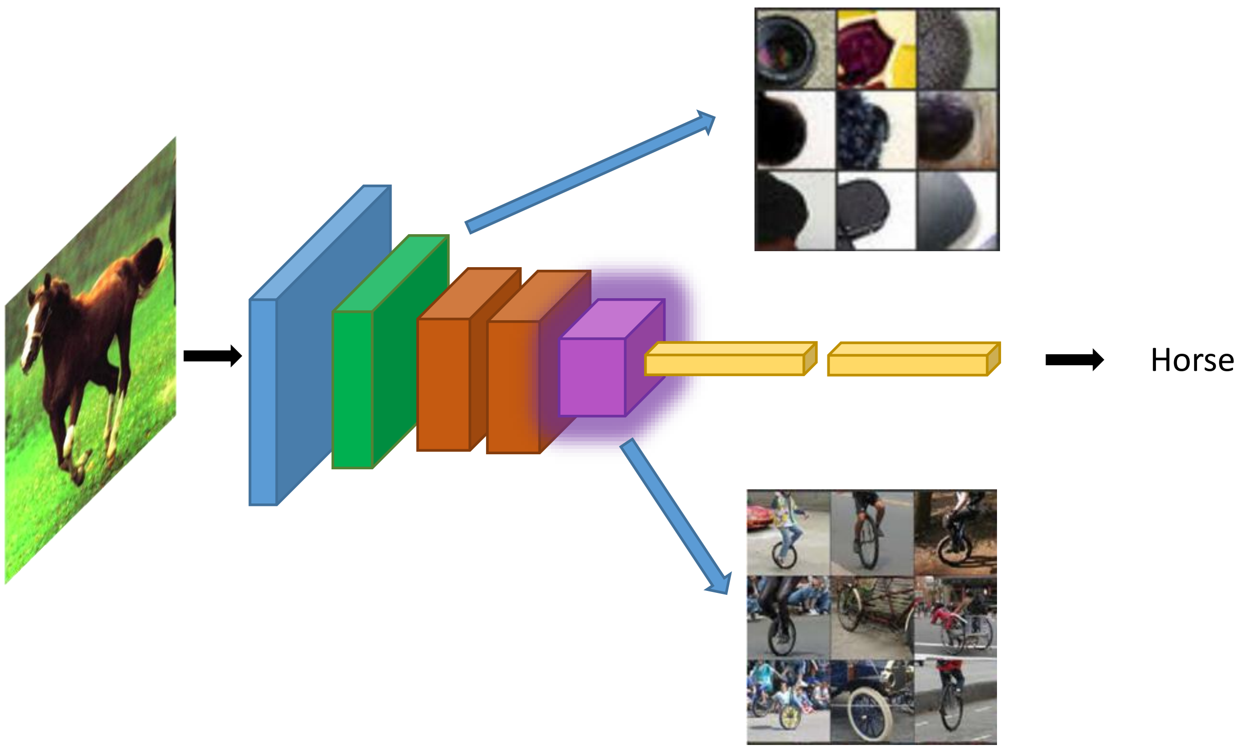
  - One-stage: YOLACT

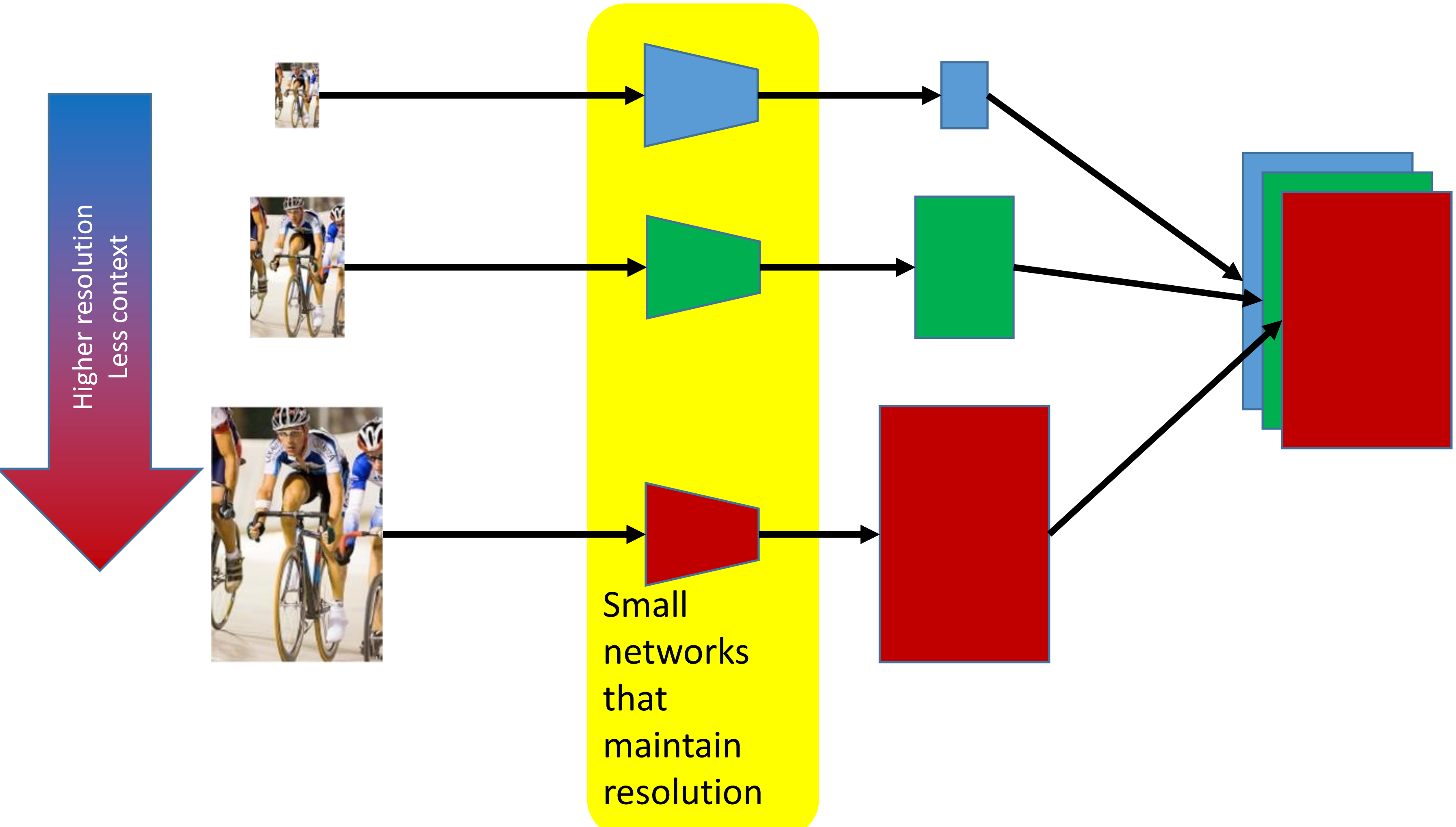
Semantic resolution problem

- Problem: early layers not semantic



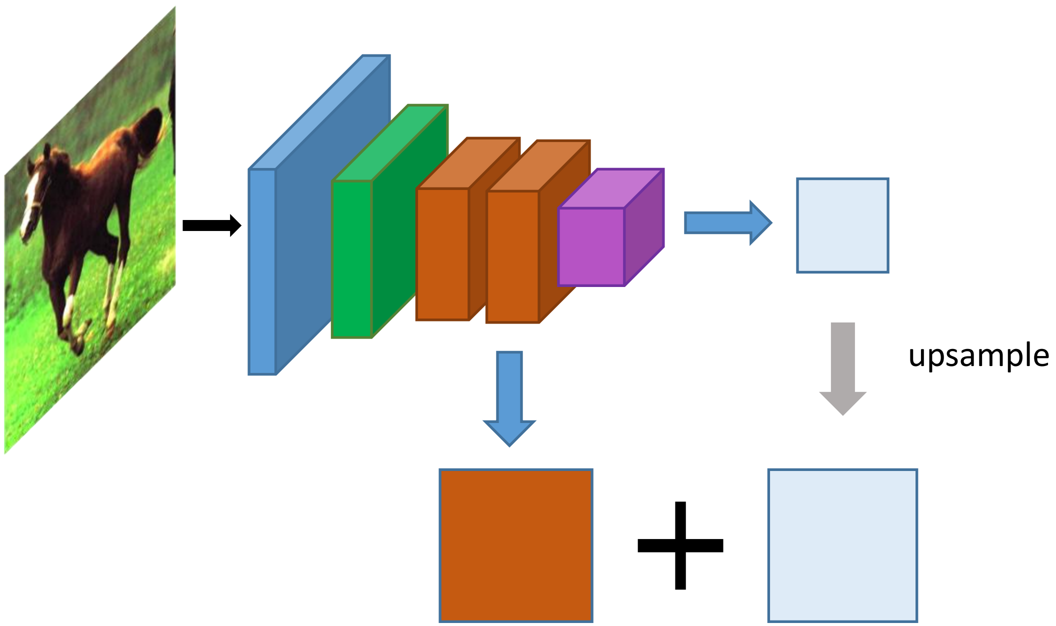
Visualizations from : M. Zeiler and R. Fergus. Visualizing and Understanding Convolutional Networks. In *ECCV* 2014.

Solution 1: Image pyramids



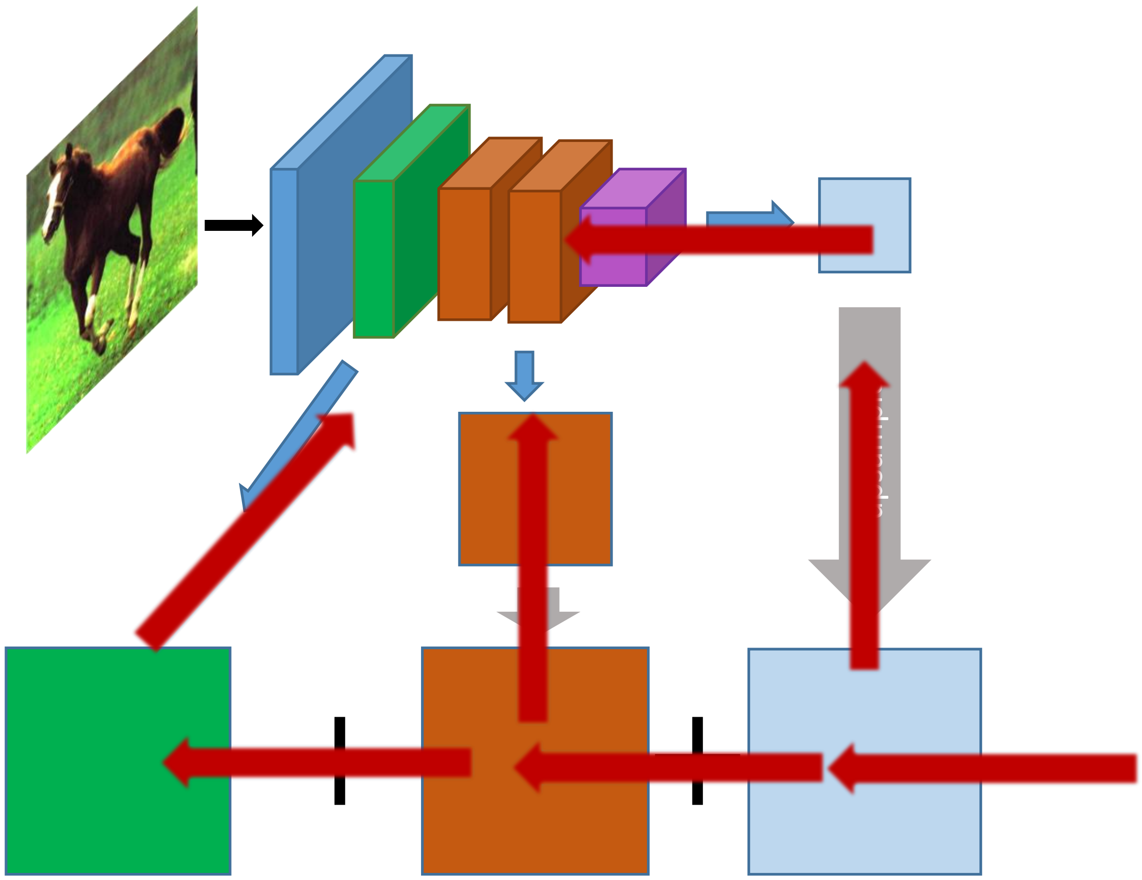
Learning Hierarchical Features for Scene Labeling. Clement Farabet, Camille Couprie, Laurent Najman, Yann LeCun. In *TPAMI,* 2013

Solution 2: Skip connections



Compute class scores at multiple layers, then upsample and add

Solution 2: Skip connections



Red arrows indicate backpropagation

Skip connections

without skip   with skip

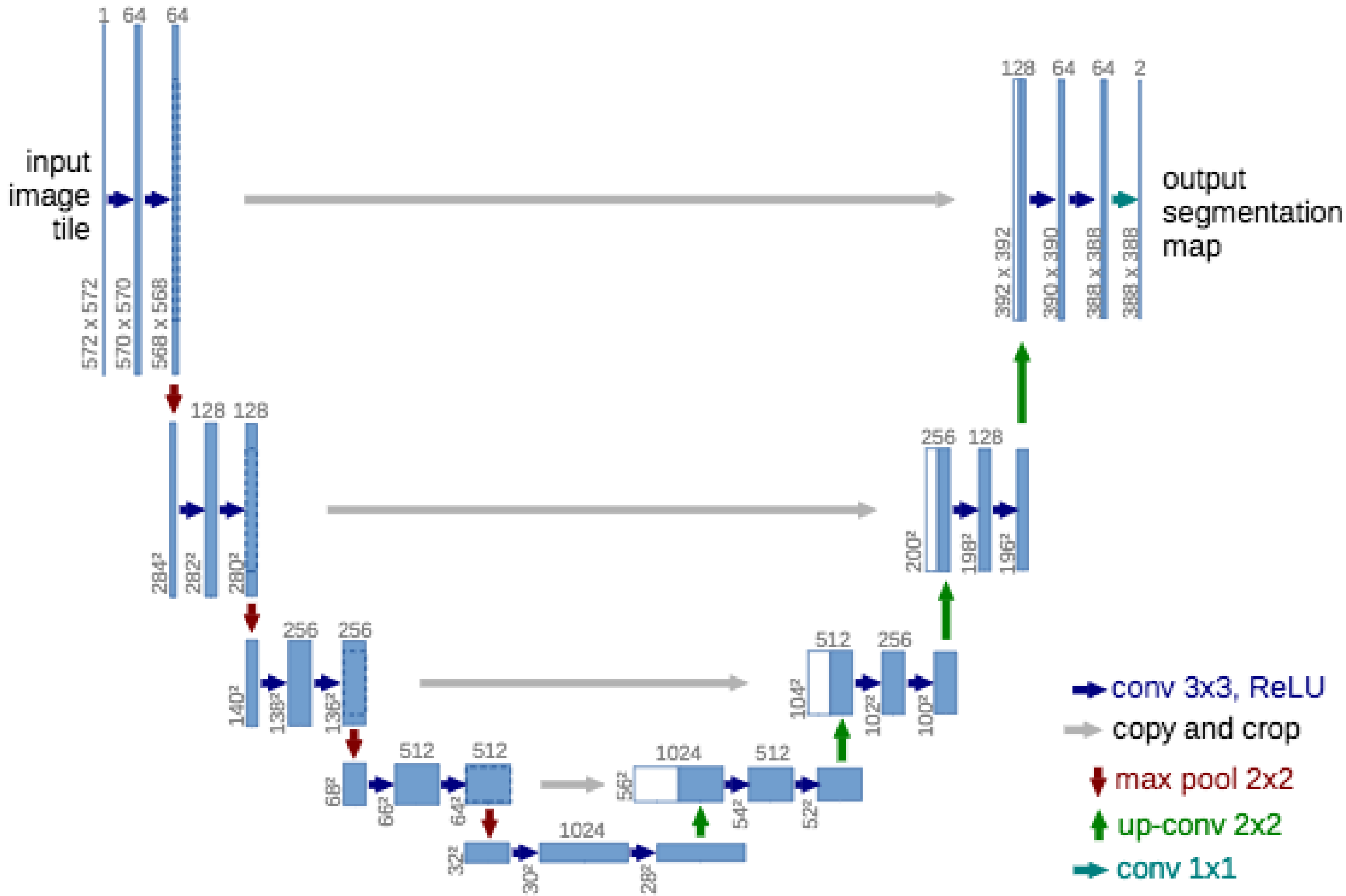
Fully convolutional networks for semantic segmentation. Evan Shelhamer, Jon Long, Trevor Darrell. In *CVPR* 2015

U-Net

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *ArXiv:1505.04597 [Cs]*. http://arxiv.org/abs/1505.04597

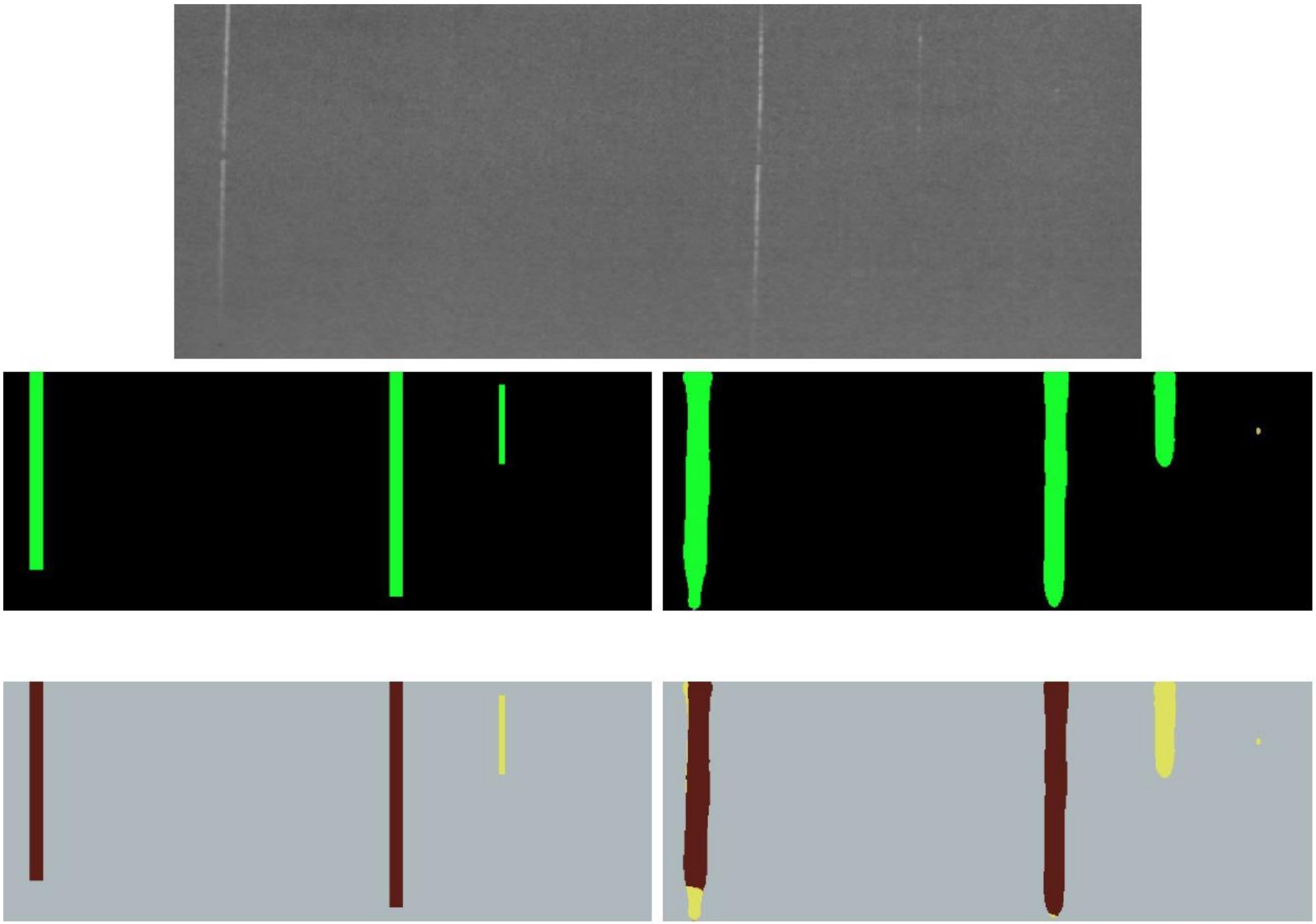
- Network for **semantic segmentation**
- Encoder-decoder-like architecture
- Decoder again contains up-conv layers
- High-resolution feature maps from encoder are combined with feature maps from decoder to improve localization

U-Net

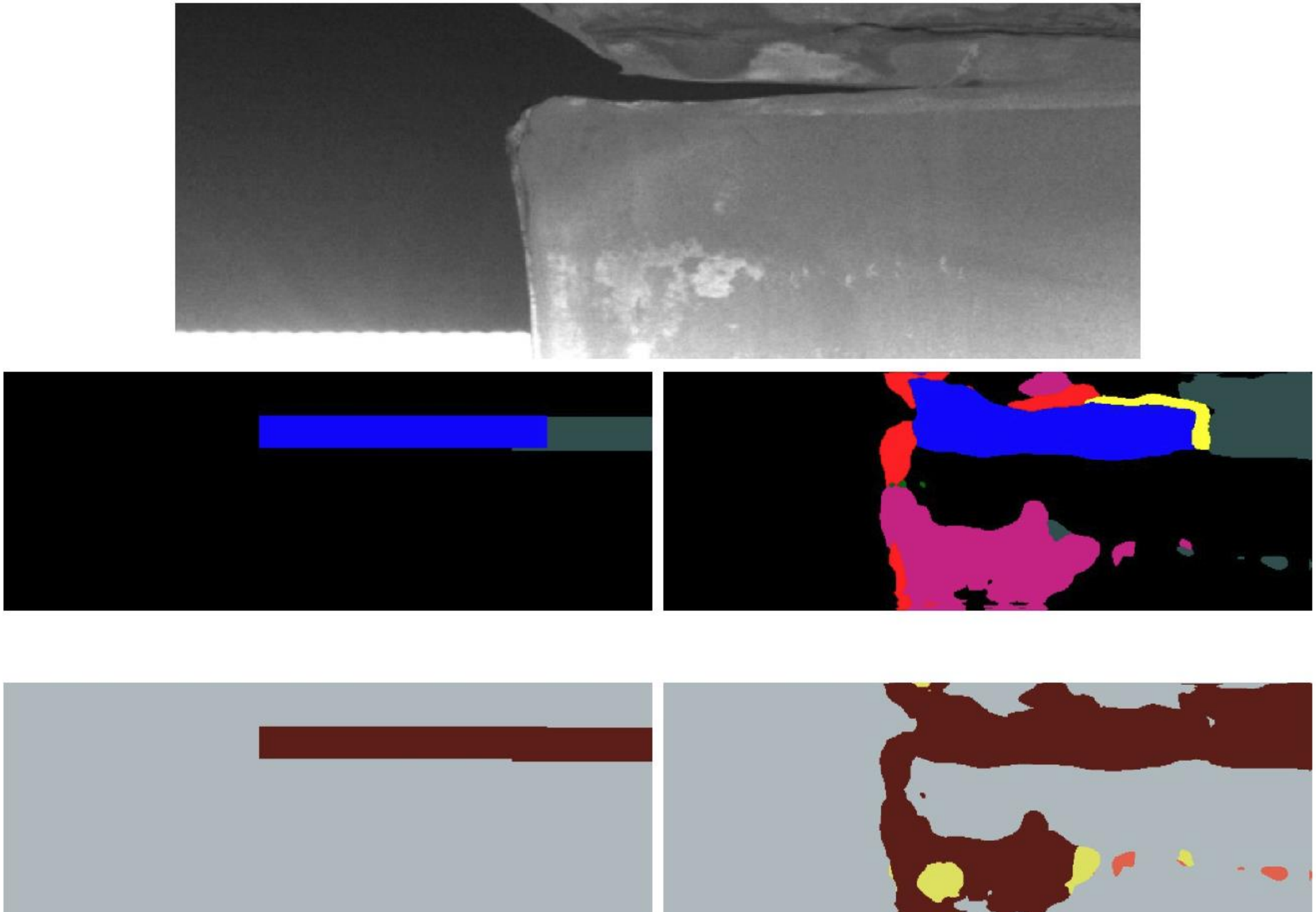


input
image
tile
output
segmentation
map
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

Example result U-Net



Example result U-Net



Semantic Image Segmentation

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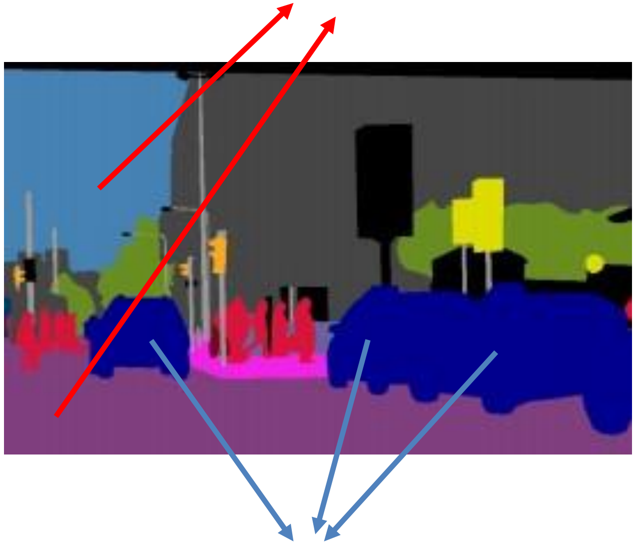
Instance segmentation

Semantic segmentation

Label every pixel, including the background (sky, grass, road)



Do not differentiate between the pixels coming from instances of the same class

Instance segmentation

Label every pixel, including the background (sky, grass, road)

Do not label pixels coming from uncountable objects (sky, grass, road)

Do not differentiate between the pixels coming from instances of the same class

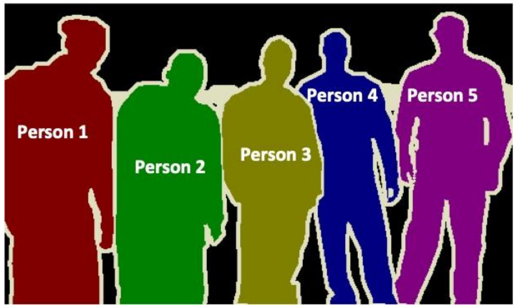
Differentiate between the pixels coming from instances of the same class

Instance segmentation methods

Proposal-based

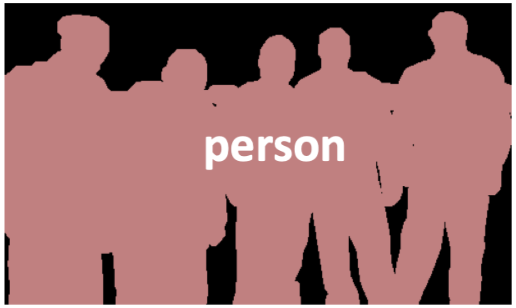
1. Proposals

2. Assign a class

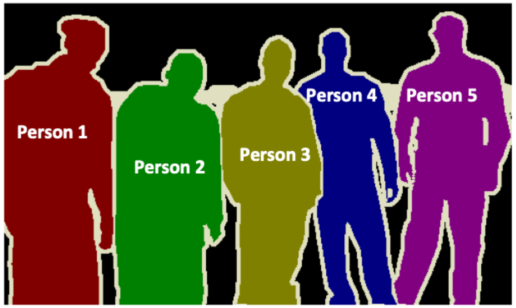


vs.

FCN-based



1. Semantic segmentation



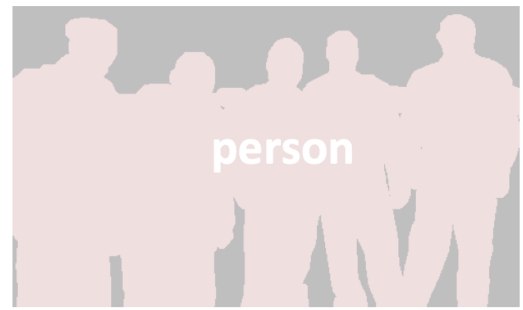
2. Find instances

Instance segmentation methods

Proposal-based

FCN-based

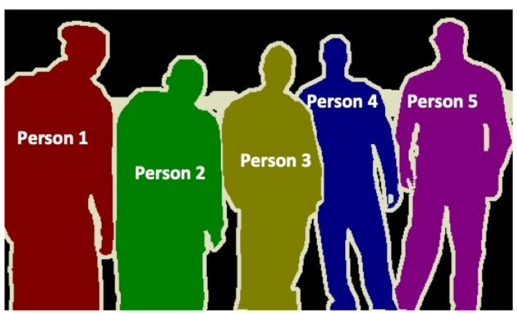
1. Proposals

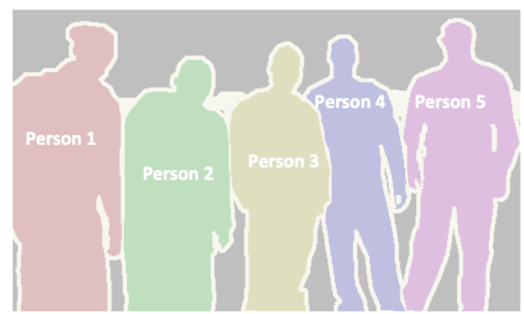


1. Semantic segmentation

vs.

2. Assign a class



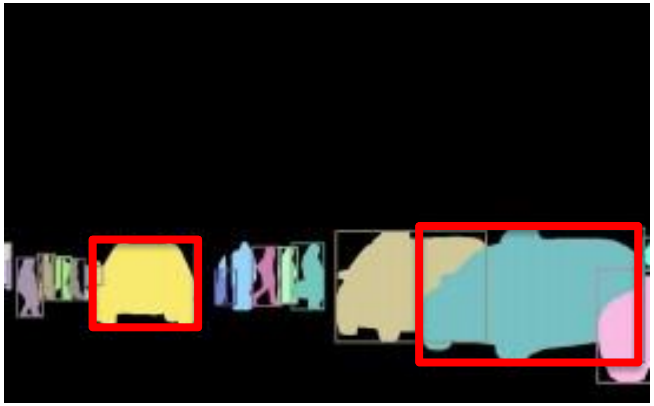


2. Find instances

Proposal-based methods

Bounding boxes.....

We already know how to obtain those!



Mask R-CNN

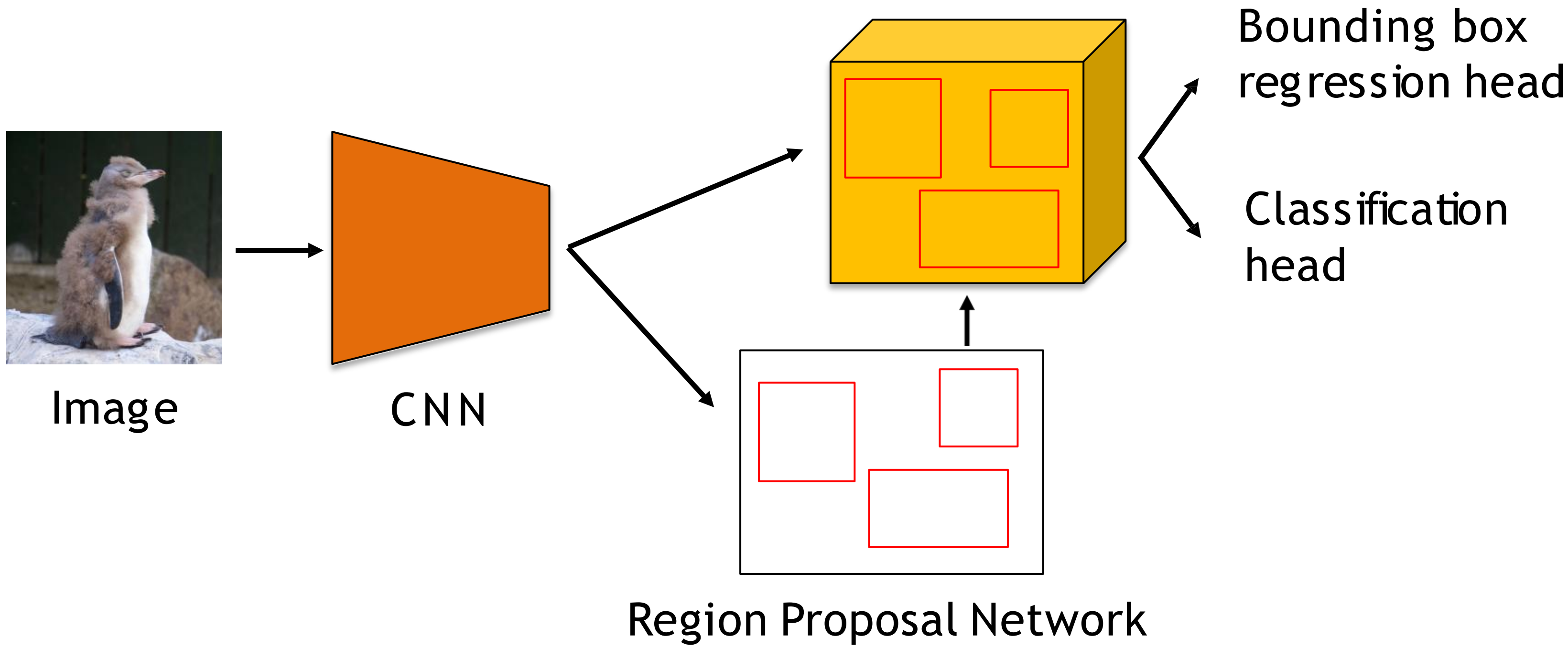
Mask R-CNN

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask R-CNN. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2961–2969. https://doi.org/10.1109/TPAMI.2018.2844175

- Network for **instance segmentation**
- Extends Faster R-CNN
- Third RoI head that outputs a 14 x 14 **binary mask** for each RoI
- Uses RoIAlign instead of RoIPool
  - Uses bilinear interpolation instead of quantization of box coordinates and when dividing into bins
  - Avoids misalignments introduced by RoIPool

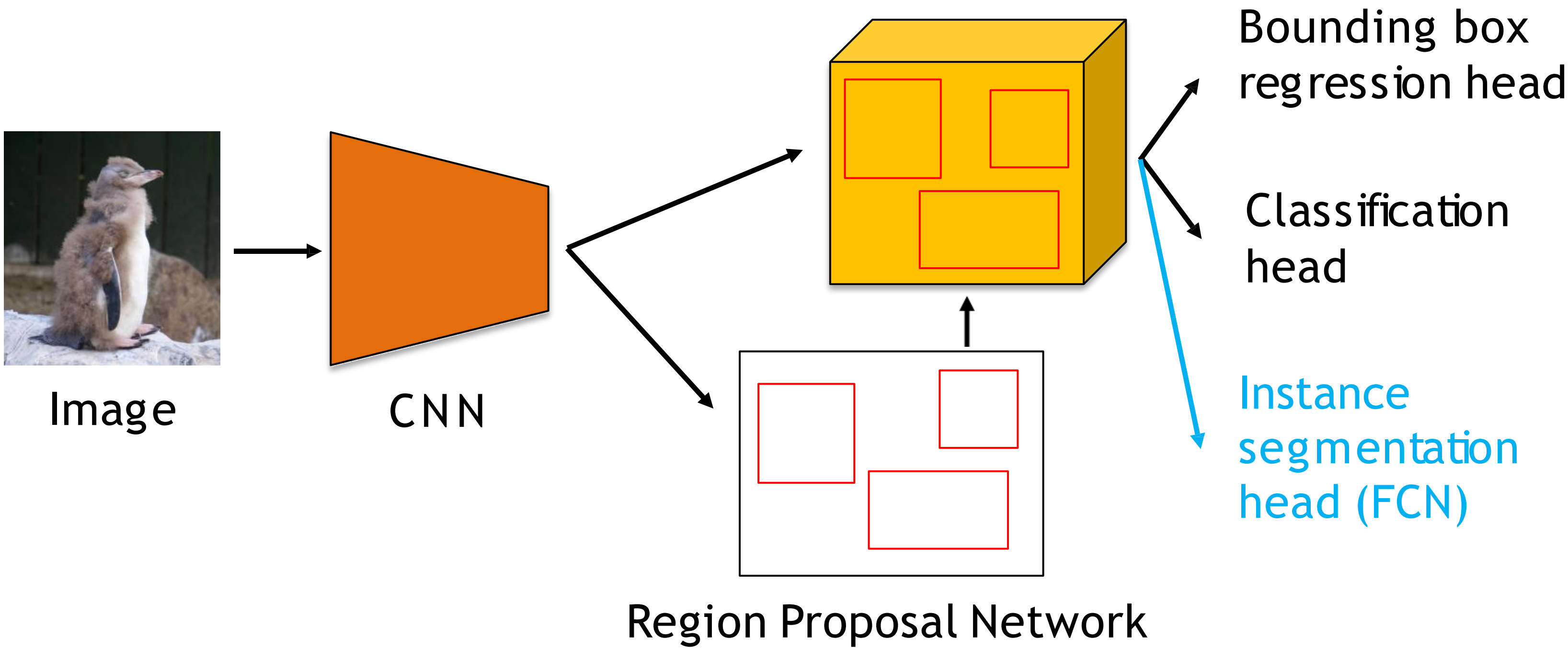
What is Mask-RCNN?

- Starting from the Faster R-CNN architecture



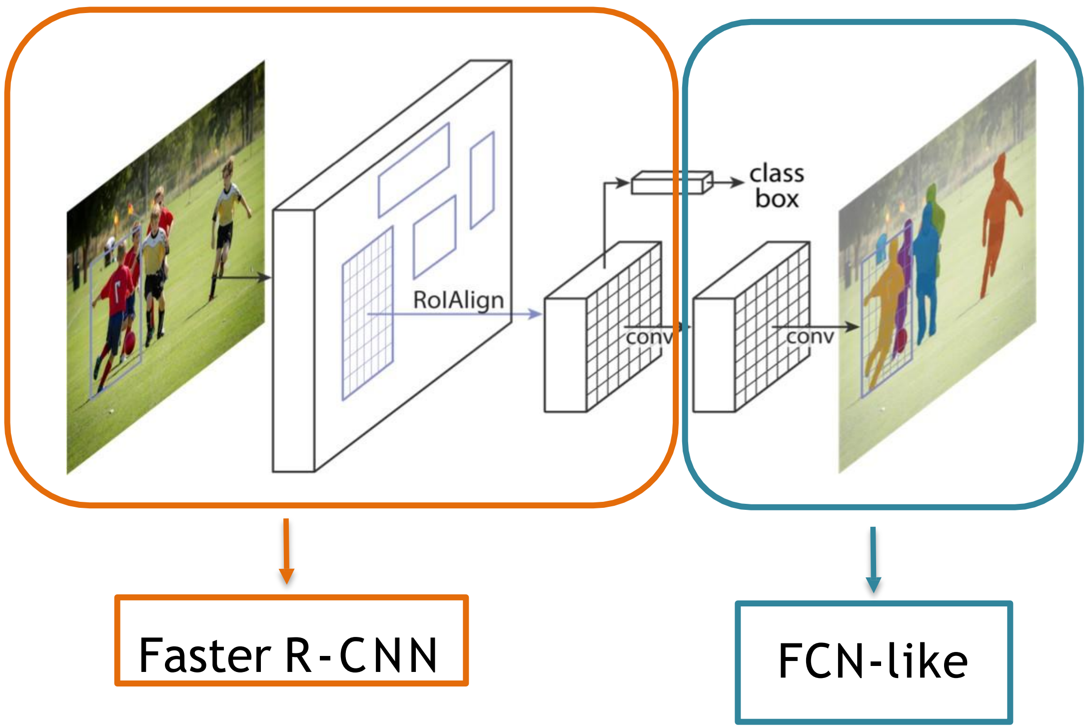
What is Mask-RCNN?

- Faster R-CNN + FCN for segmentation



What is Mask-RCNN?

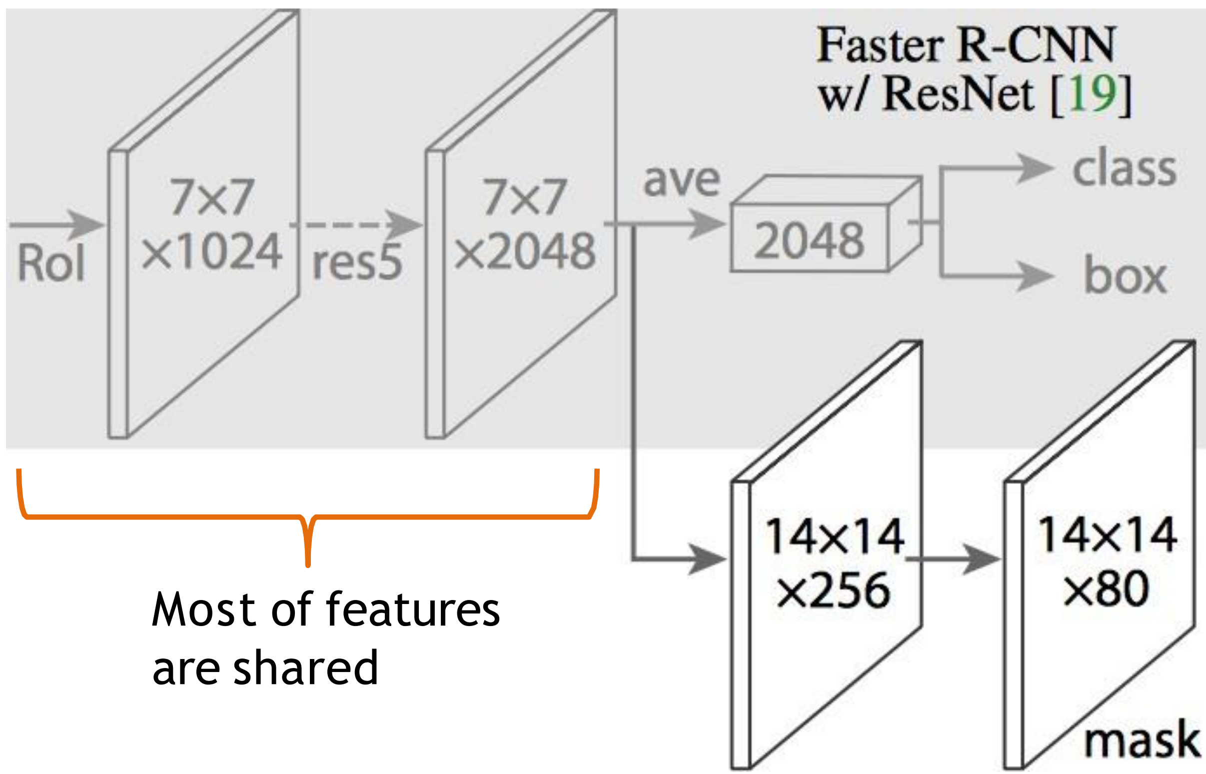
- Faster R-CNN + FCN for segmentation



Mask loss =
binary cross
entropy per pixel
for the k semantic
classes

He at al. 'Mask R-CNN" ICCV 2017

Mask R-CNN



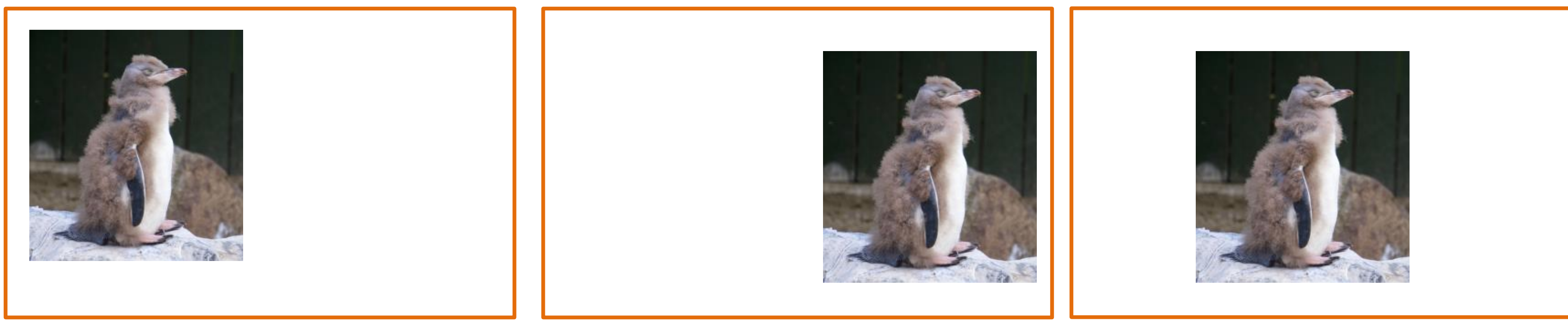
Object recognition head

Segmentation head

Most of features are shared

Detection vs. segmentation

- Detection: for object classification, you require invariant representations



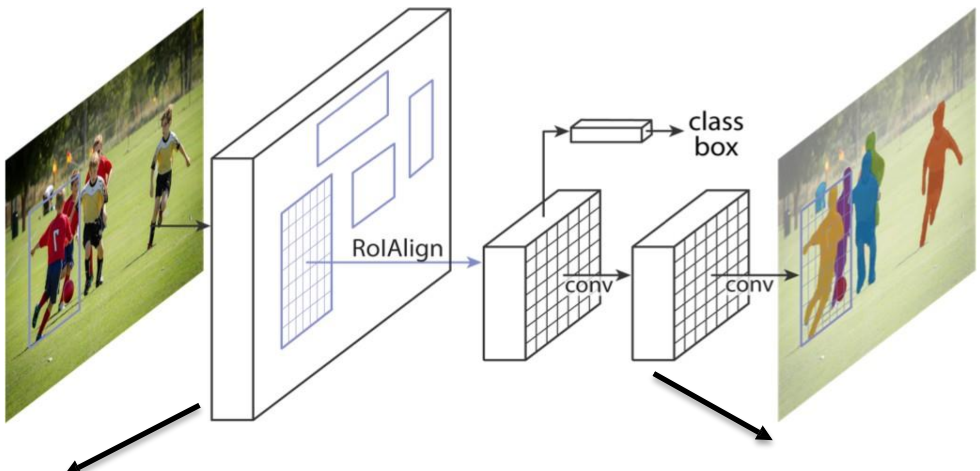
Translation invariance: wherever the penguin is in the image, I still want to have ‘penguin”as my classification output

Detection vs. segmentation

- Detection: for object classification, you require invariant representations
- Segmentation: you require equivariant representations
  - Translated object → Translated mask
  - Scaled object → scaled mask
  - For semantic segmentation, small objects are less important (less pixels), but for instance segmentation, all objects (no matter the size) are equally important

Mask-RCNN: operations

- What operations are equivariant?



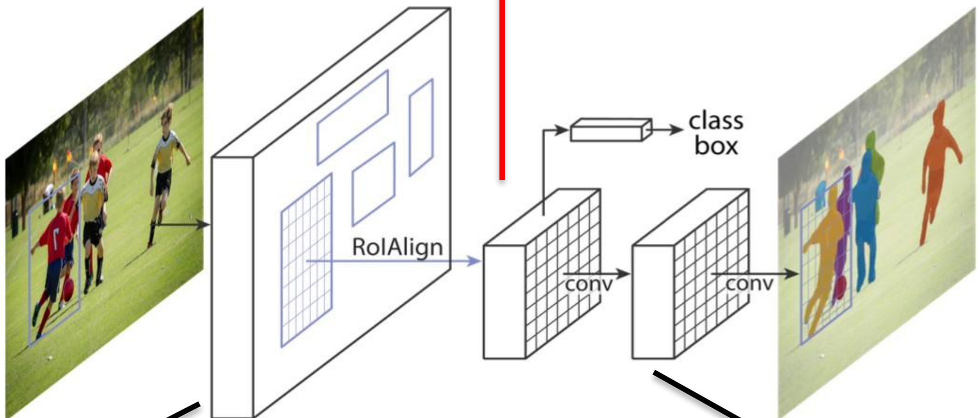
Features extraction = convolutional layers → equivariant

Segmentation head is a fully convolutional network → equivariant

Mask-RCNN: operations

- What operations are equivariant?

Fully connected layers and global pooling layers give invariance!

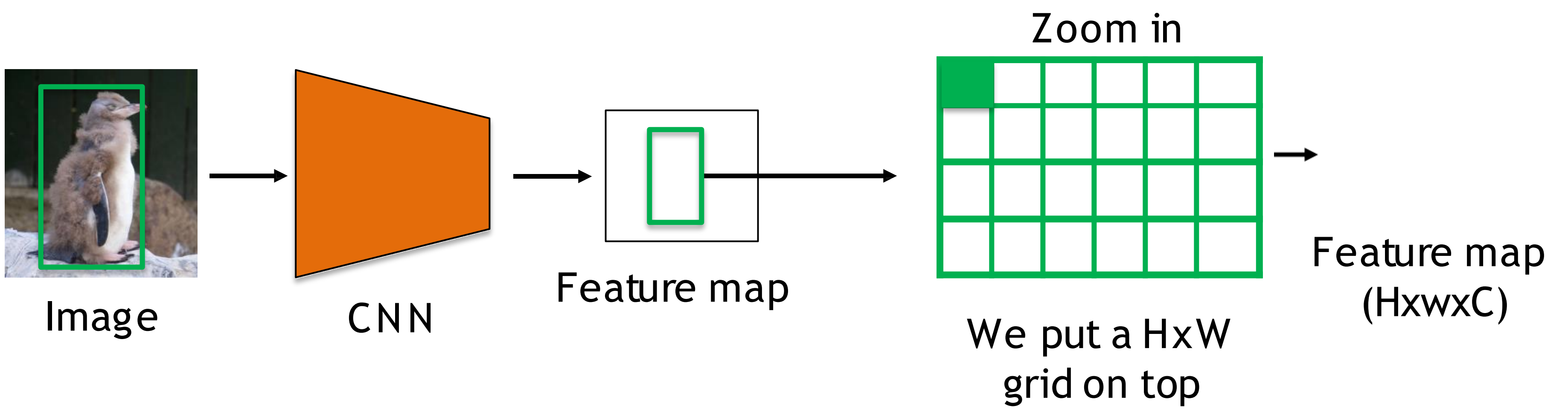


Features extraction = convolutional layers → equivariant

Segmentation head is a fully convolutional network → equivariant

Recall: RoI pooling

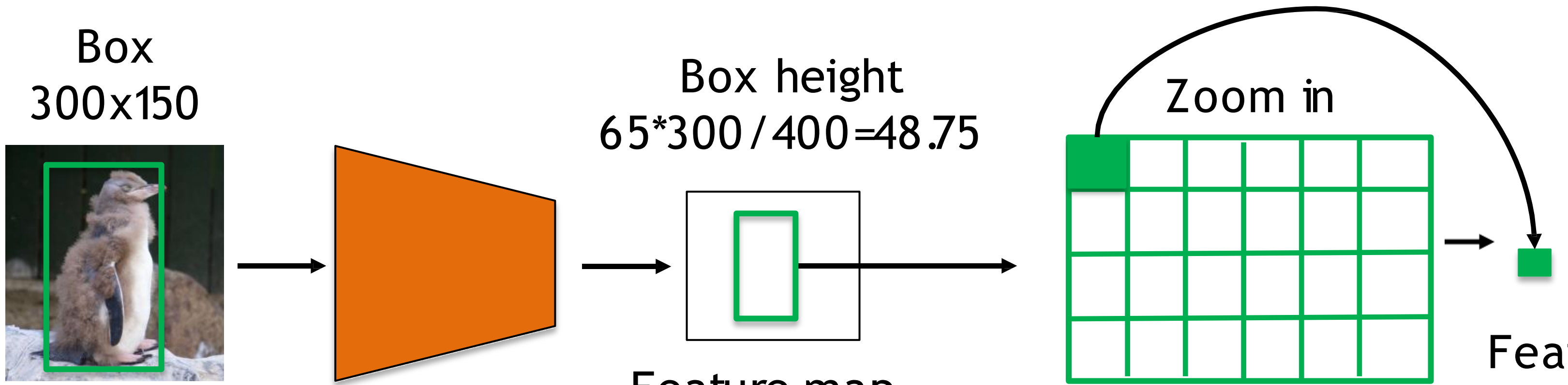
- Region of Interest Pooling: for every proposal



Recall: RoI pooling

Not suitable to extract pixel-wise precise masks

- Let us look at sizes



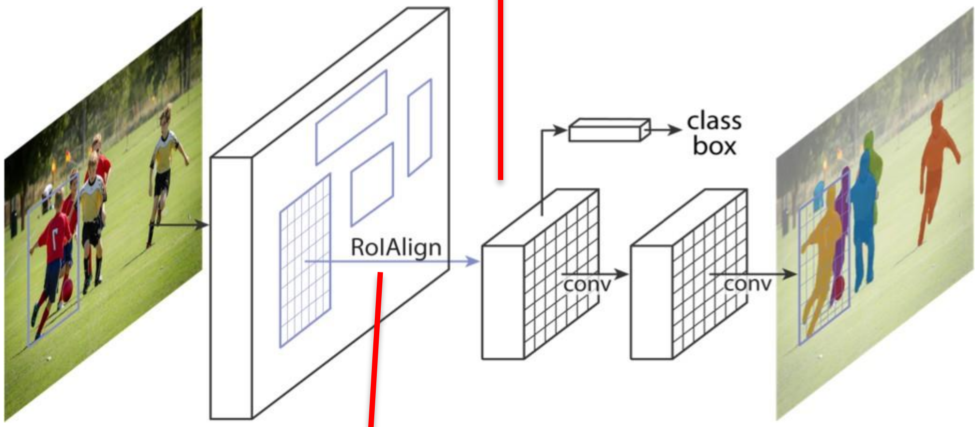
Quantization effect = chose 48

Quantization effect

Mask-RCNN: operations

- Make all operations equivariant

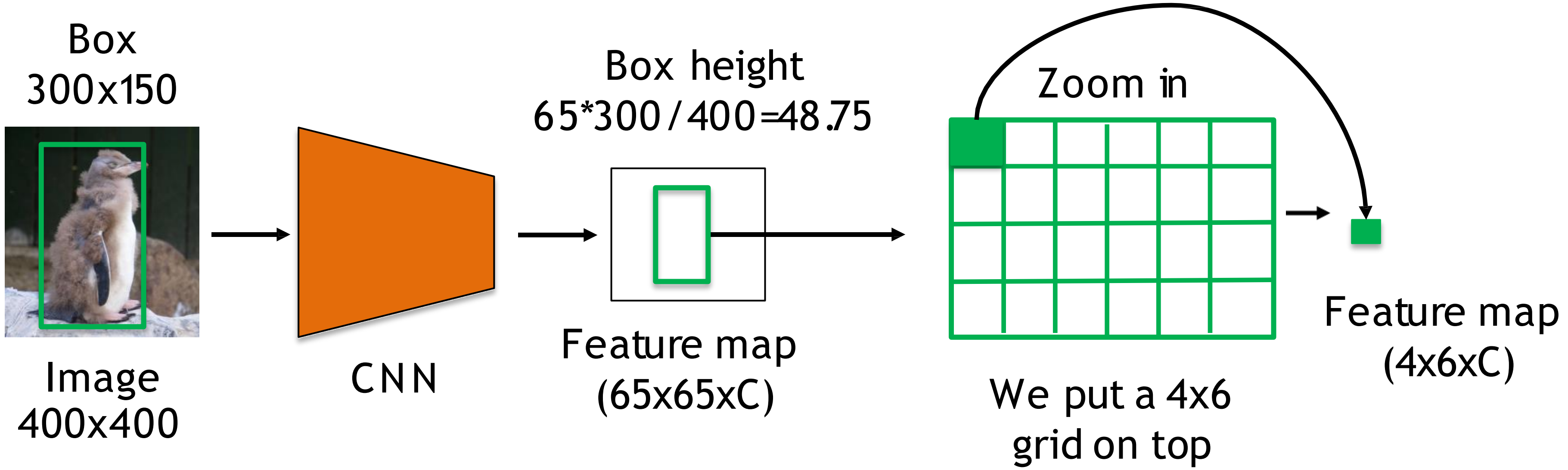
Fully connected layers and global pooling layers give invariance!



Exchange RoI pooling by an equivariant operation = RoI Align

RoIAlign

- Erase quantization effects



Chose 48.75

ROIAlign

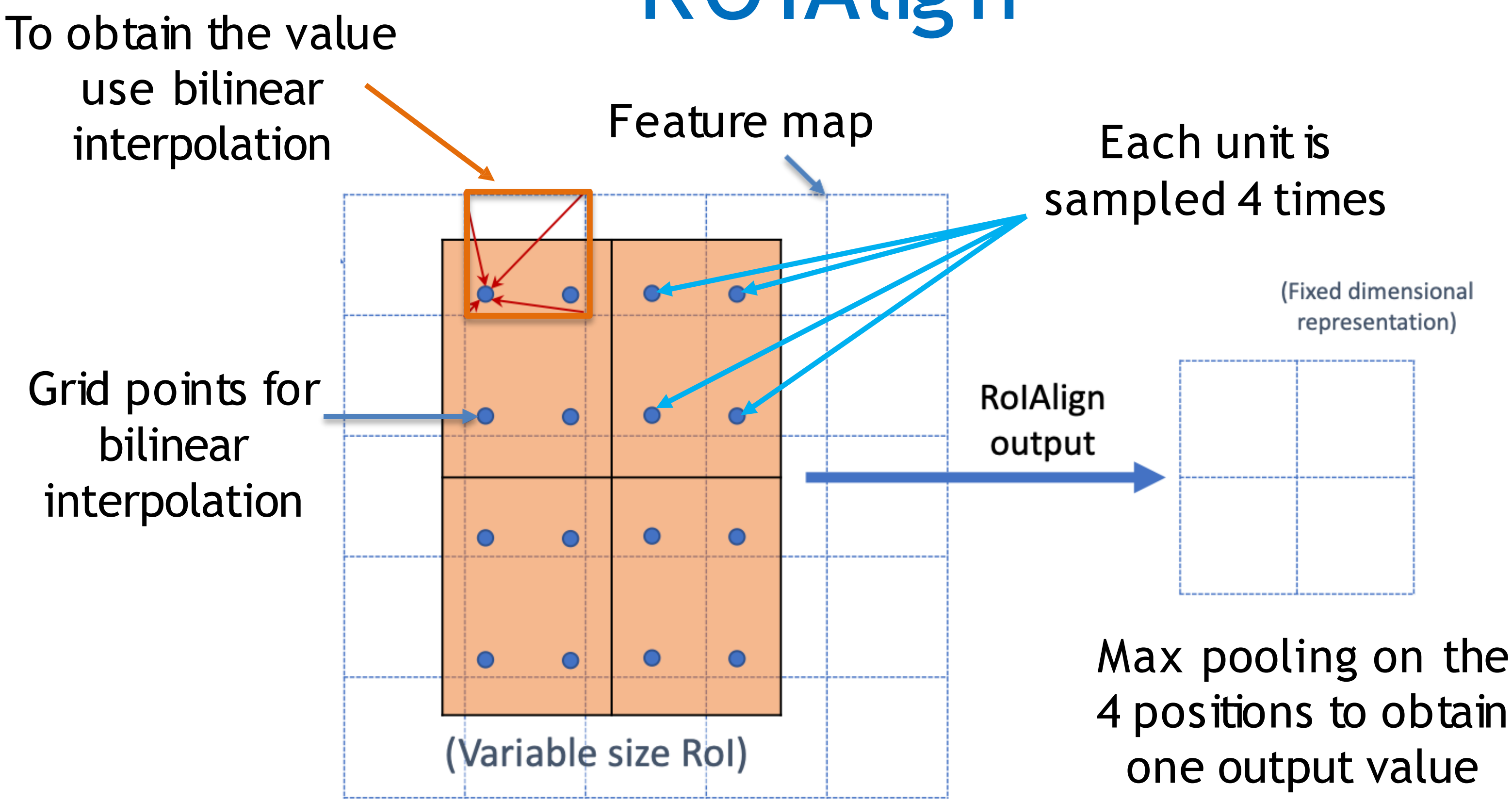
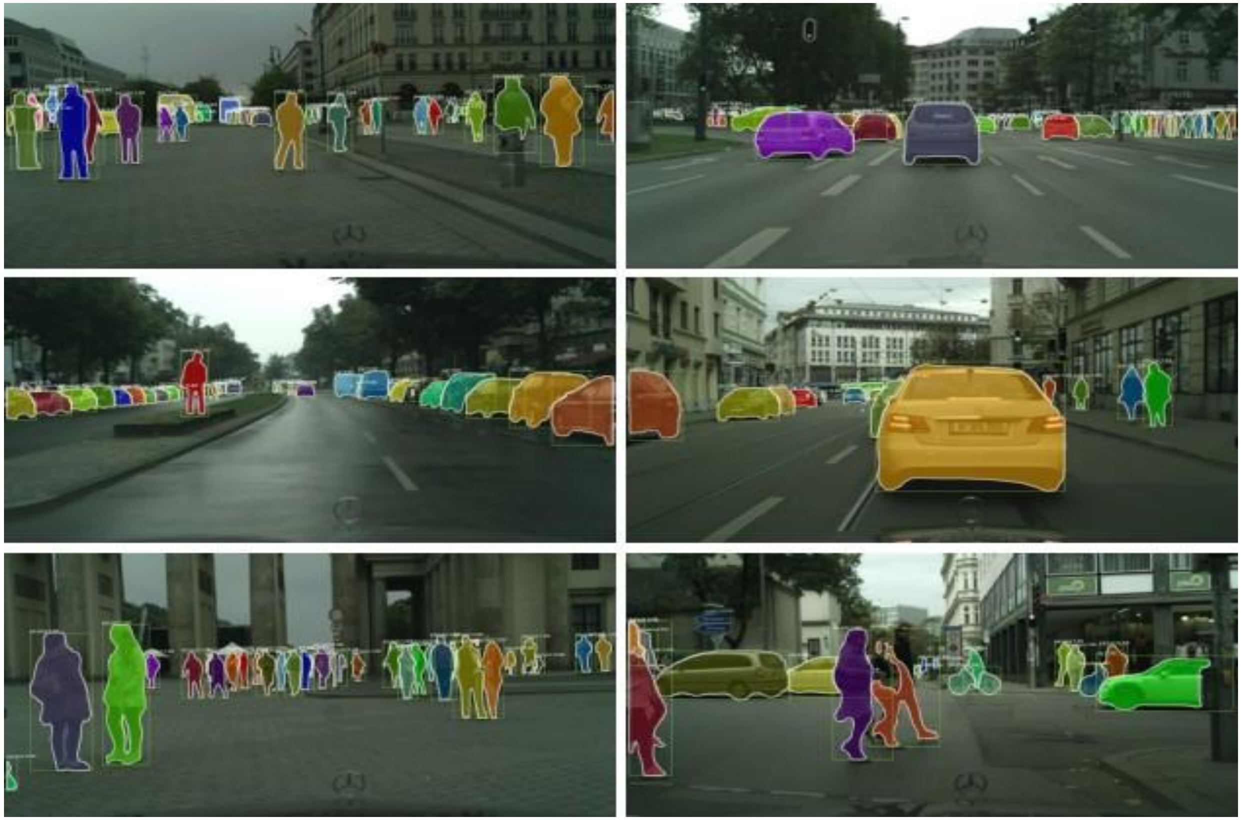


Image: Kaiming He

Mask R-CNN: qualitative results

Mask R-CNN: qualitative results

Mask R-CNN: qualitative results



Mask R-CNN: extended for joints

Model a keypoint's location as a one-hot mask, and adopt Mask R-CNN to predict K masks (which are in the end only 1pixel), one for each of K keypoint types (e.g., left shoulder, right elbow). This demonstrates the flexibility of Mask R-CNN.

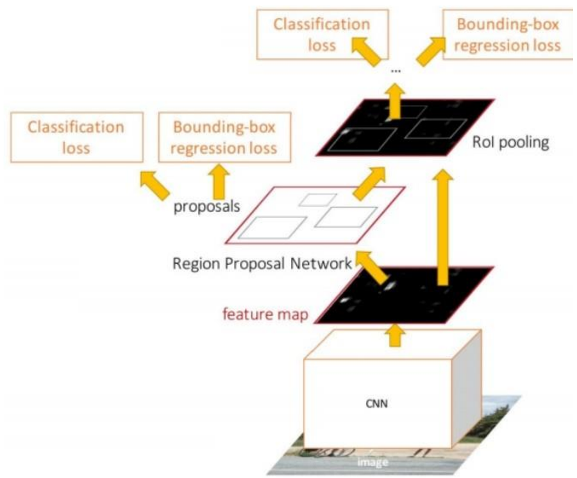
Semantic Image Segmentation

- Semantic segmentation:
  - Classic techniques using colour
  - Fully Convolutional Network
  - Deconvolution
  - Skip connections: U-Net
- Instance Segmentation
  - Two-stage: Mask-RCNN
  - One-stage: YOLACT

Is one-stage vs two-stage also applicable to masks?

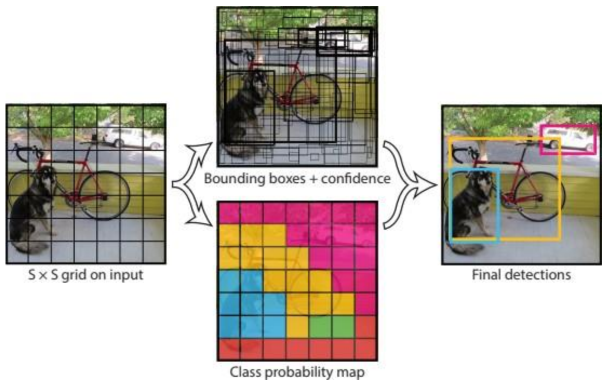
One-stage vs two-stage detectors

Faster R-CNN



Slower, but has higher performance

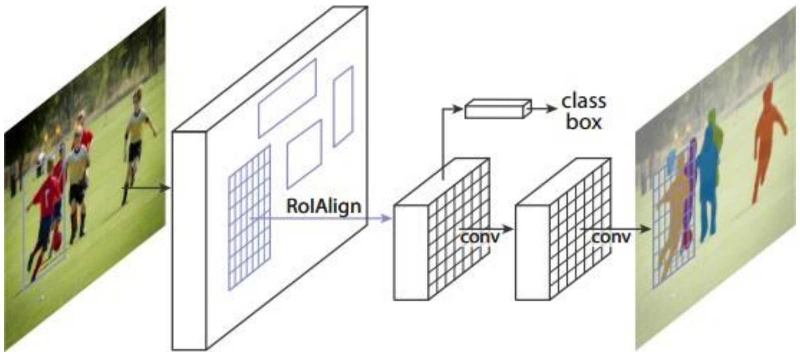
YOLO



Faster, but has lower performance

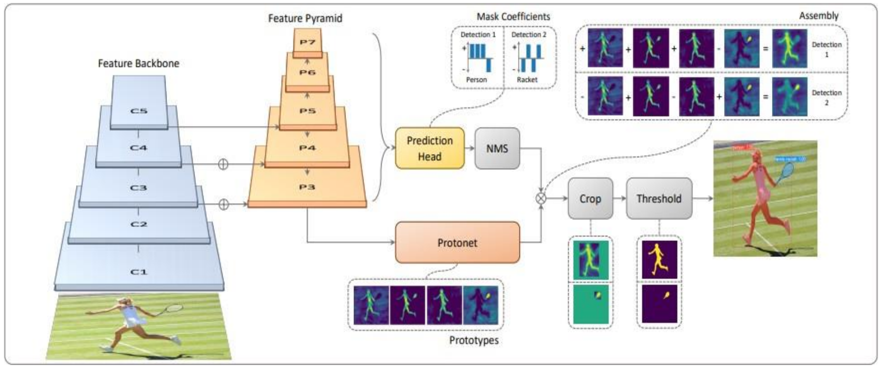
One-stage vs two-stage instance segmenters

Mask R-CNN



Slower, but has higher performance

YOLACT



Faster, but has lower performance

YOLO with masks?

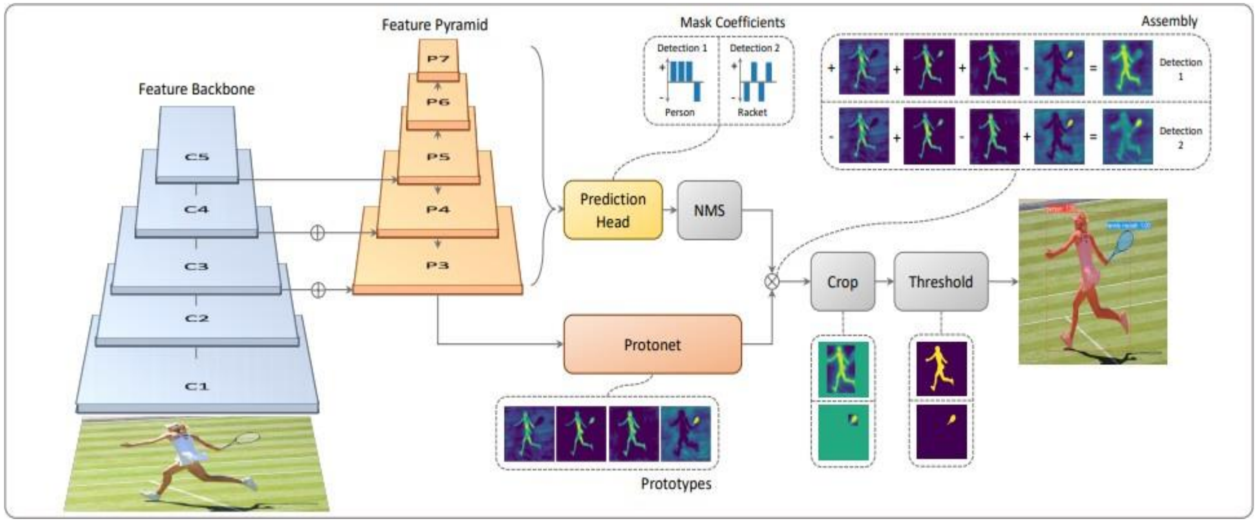
"Boxes are stupid anyway though, I'm probably a true believer in masks except I can't get YOLO to learn them."

– Joseph Redmon, YOLOv3

YolACT*

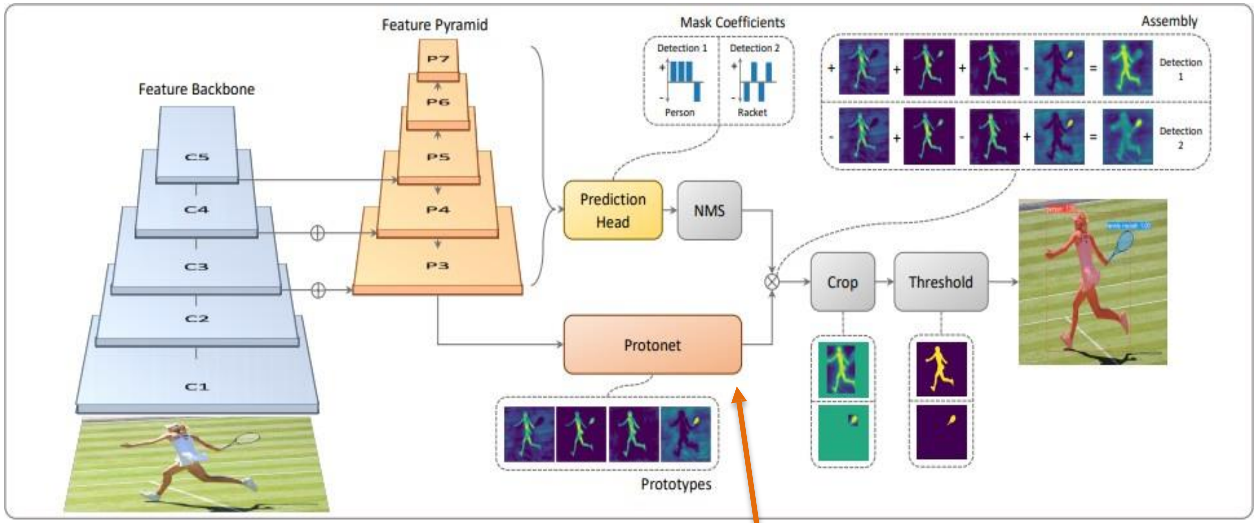
*You Only Look At CoefficienTs

YOLACT: idea



D. Bolya et al. “YOLACT: Real-time Instance Segmentation”. ICCV 2019

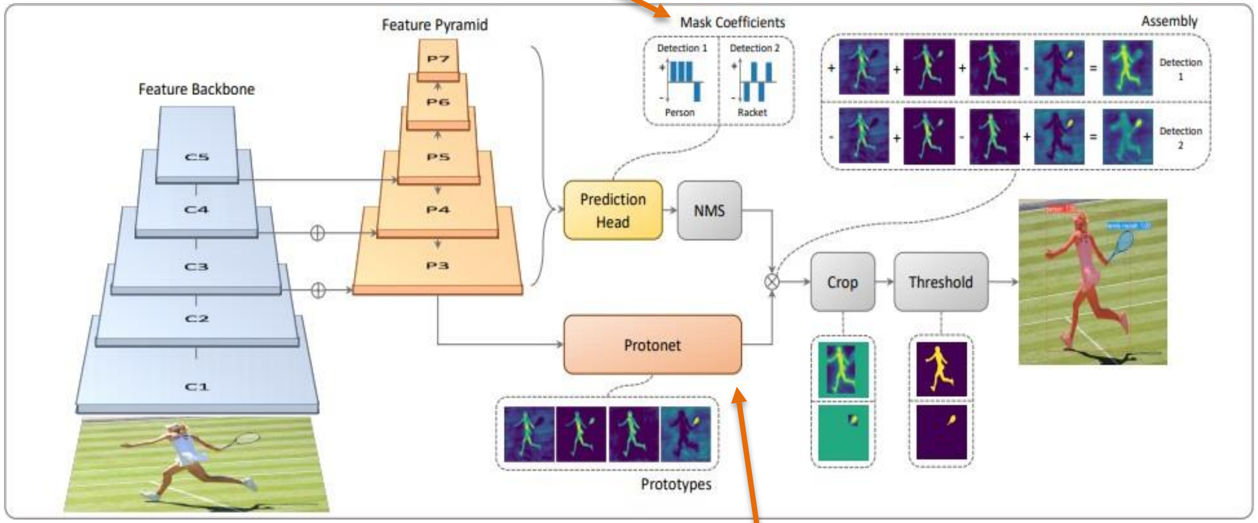
YOLACT: idea



1) Generate mask prototypes

YOLACT: idea

2) Generate mask coefficients

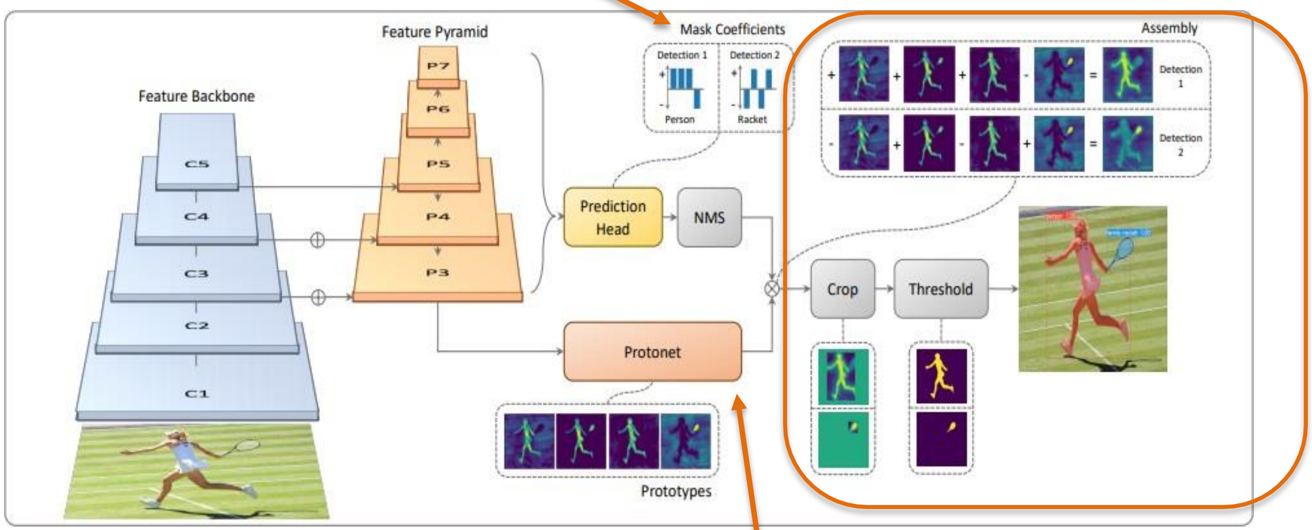


1) Generate mask prototypes

YOLACT: idea

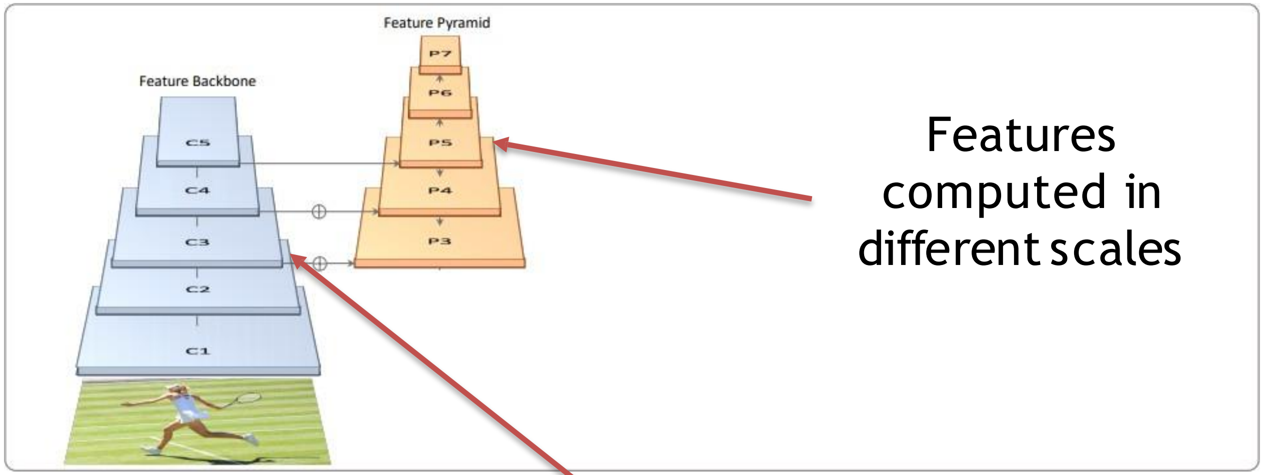
2) Generate mask coefficients

3) Combine (1) and (2)



1) Generate mask prototypes

YOLACT: backbone

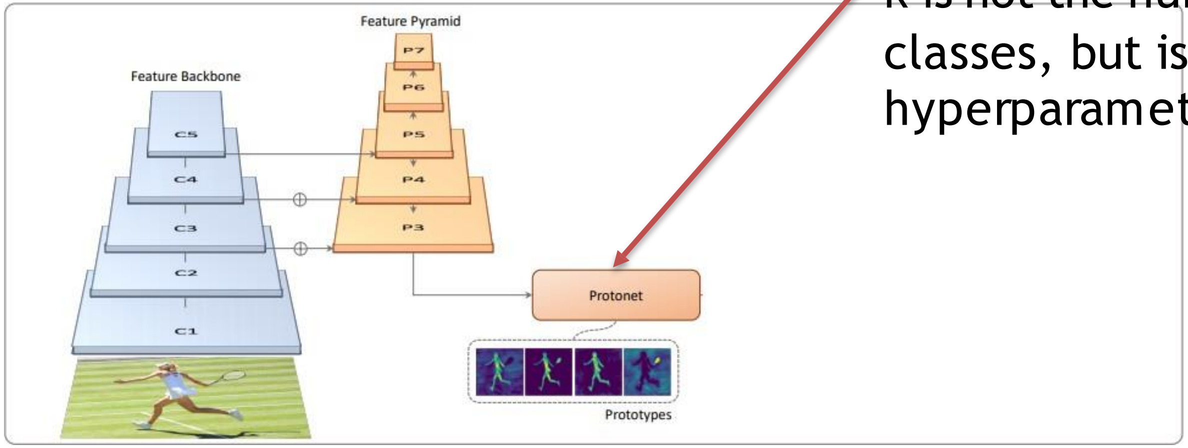


Features computed in different scales

ResNet-101

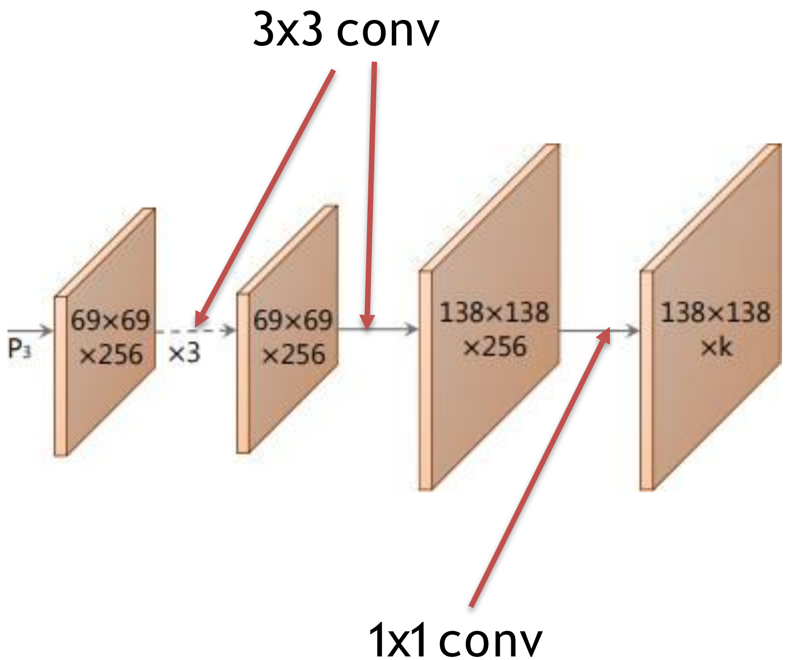
YOLACT: protonet

Generate k prototype masks.
k is not the number of classes, but is a hyperparameter.



YOLACT: protonet

- Fully convolutional network

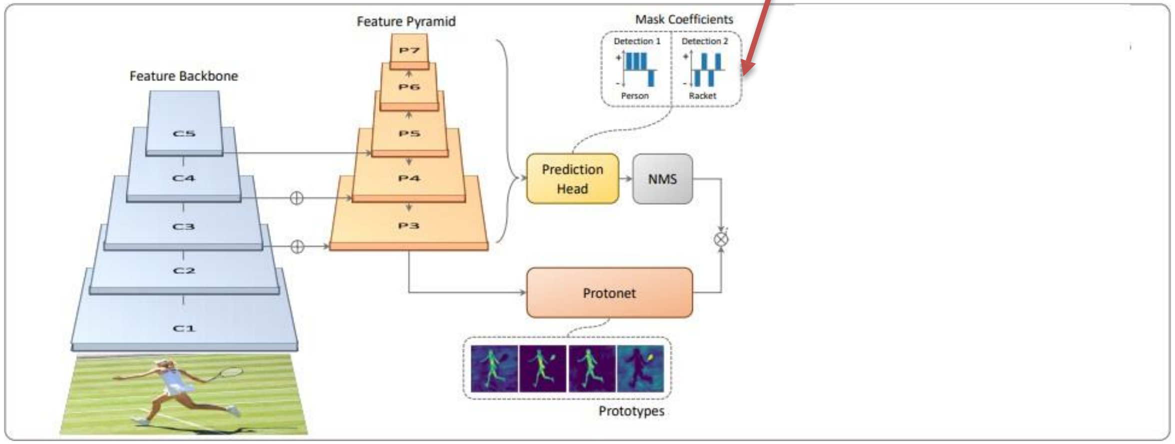


Similar to the mask branch in Mask R-CNN.

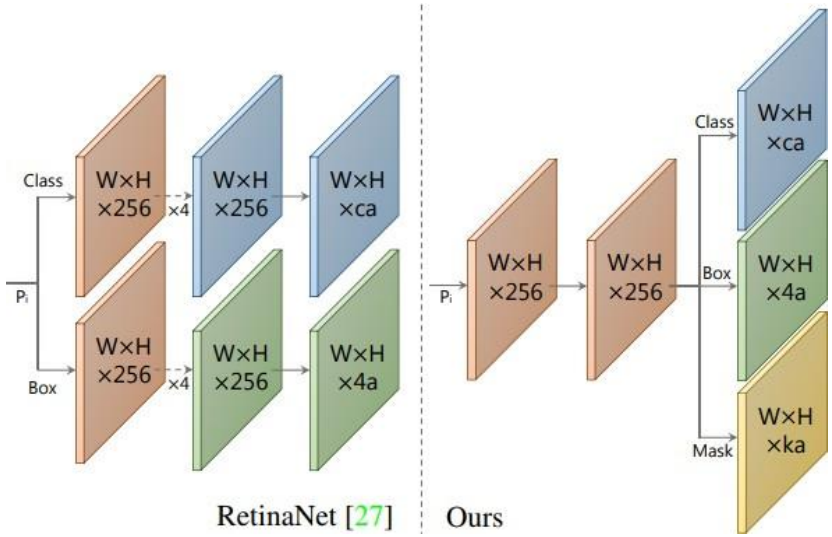
However, no loss function is applied on this stage.

YOLACT: mask coefficients

Predict a coefficient for every predicted mask.



YOLACT: mask coefficients



Predict one class per anchor box

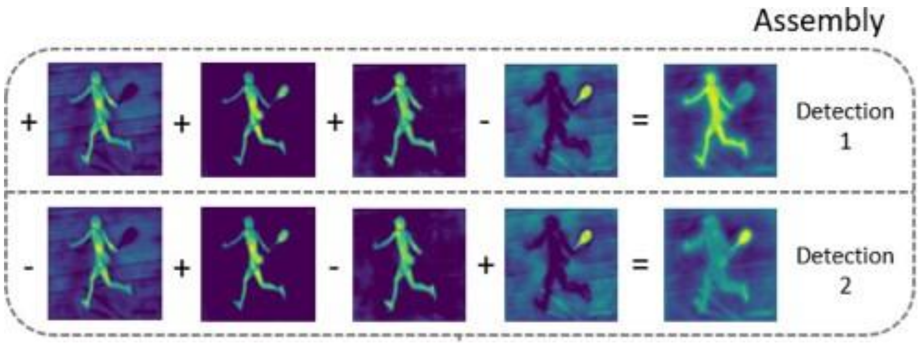
Predict the regression per anchor box

Predict k coefficients (one per prototype mask) per anchor

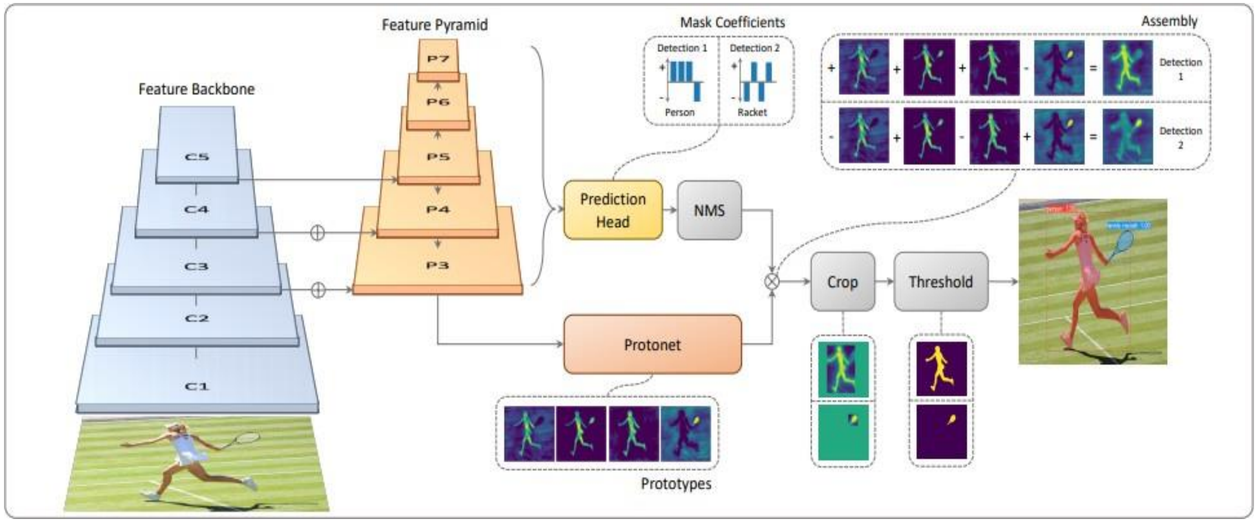
The network is similar but shallower than RetinaNet

YOLACT: mask assembly

1. Do a linear combination between the mask coefficients and the mask prototypes.
2. Predict the mask as $M = \sigma(PC^T)$ where P is a (HxWxK) matrix of prototype masks, C is a (NxK) matrix of mask coefficients surviving NMS, and σ is a nonlinearity.



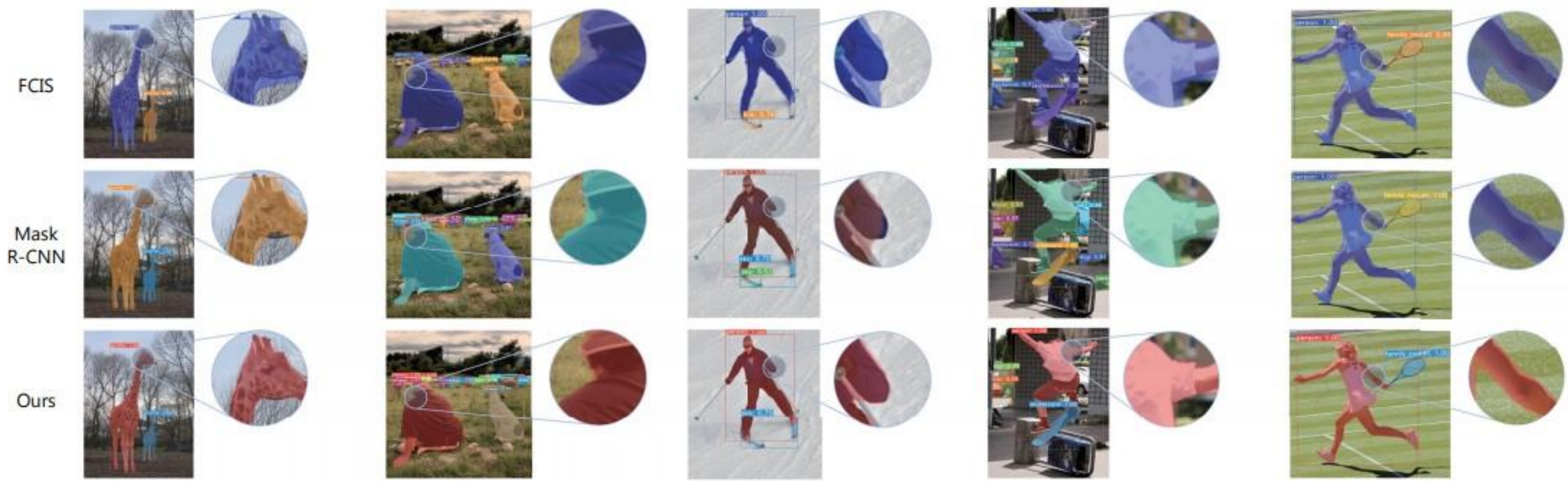
YOLACT: loss function



Cross-entropy between the assembled masks and the ground truth, in addition to the standard losses (regression for the bounding box, and classification for the class of the object/mask).

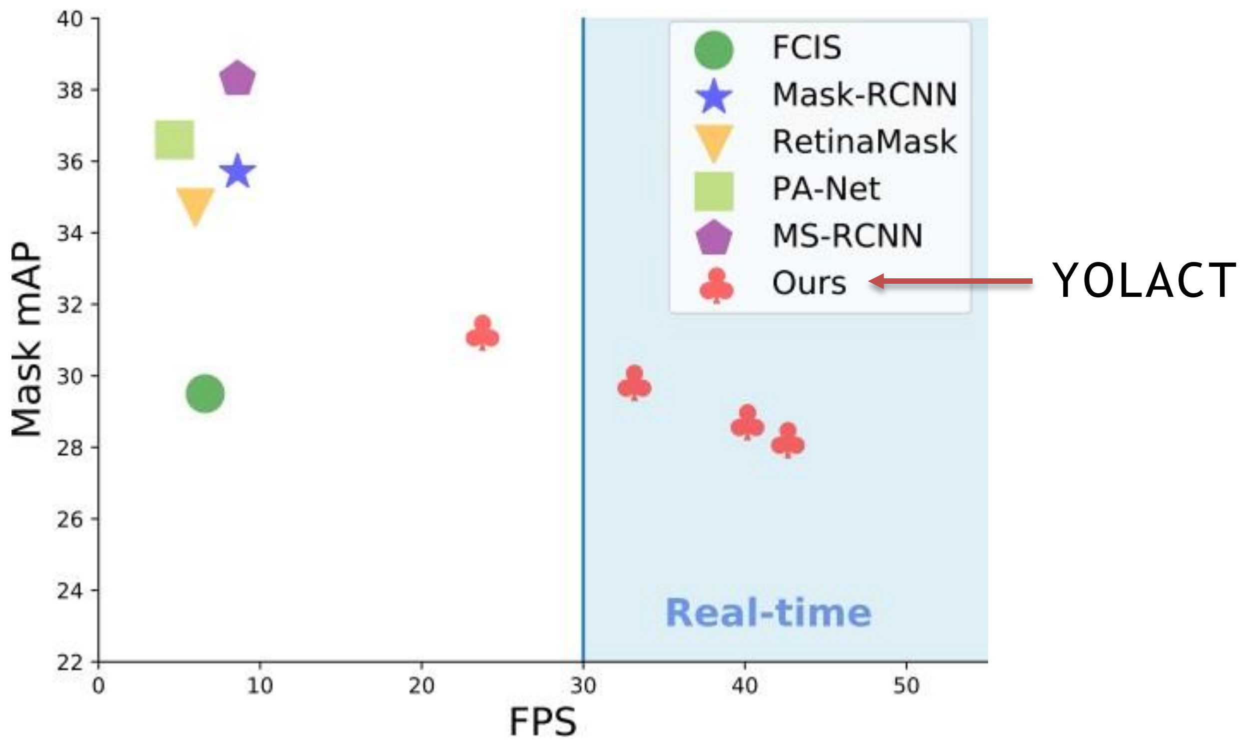
YOLACT: qualitative results

YOLACT: qualitative results



For large objects, the quality of the masks is even better than those of two-stage detectors

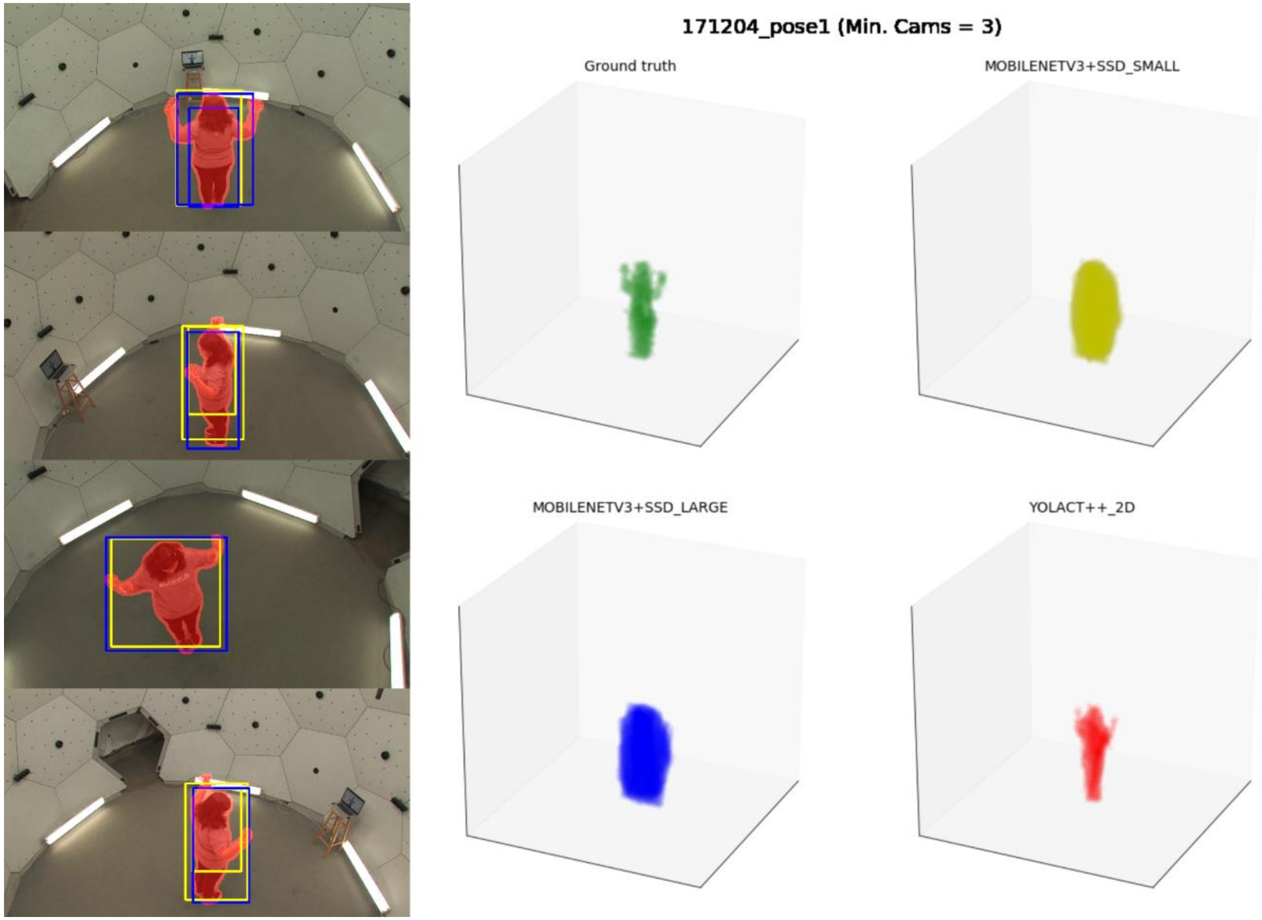
So, which segmenter to use?



YOLACT: improvements

- A specially designed version of NMS, in order to make the procedure faster.
- An auxiliary semantic segmentation loss function performed on the final features of the FPN.
The module is not used during the inference stage.
- D. Boyla et al. "YOLACT++: Better real-time instance segmentation". arXiv:1912.06218 2019

Example result Instance segmentation (YOLACT++)



https://www.youtube.com/watch?v=rQh2RpEQyKI

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